

Electromagnetism at low frequencies

BIOELECTRICITY, NEURAL MODELS AND BIO-TISSUES STIMULATION

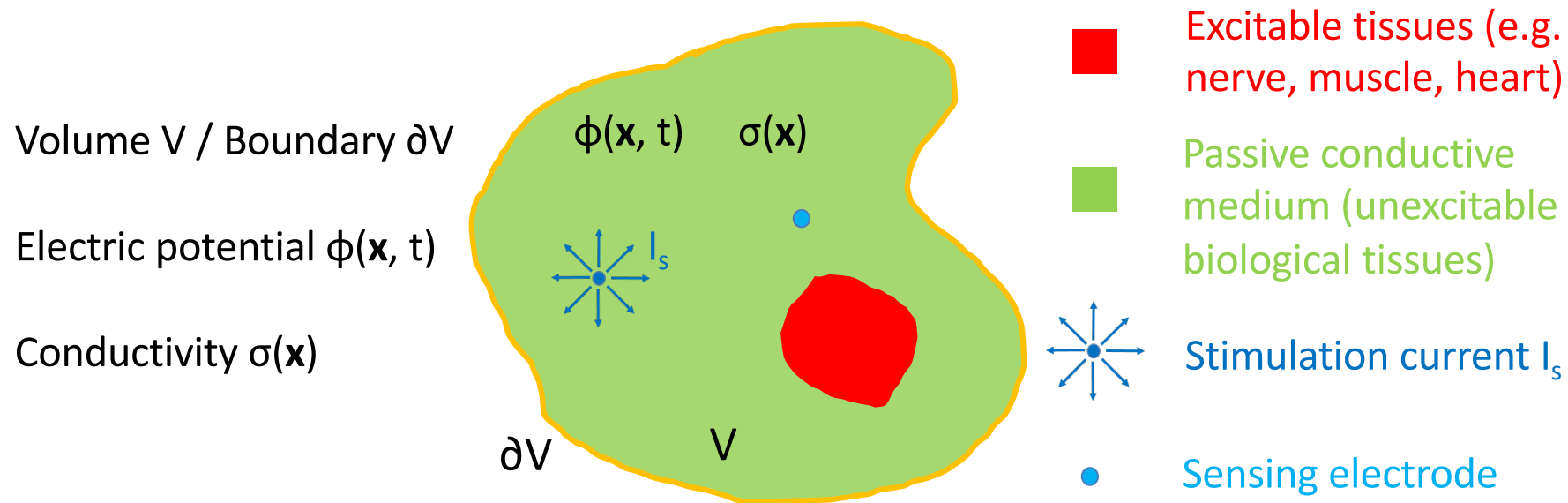
A.A. 2025-26

Prof. Alessandro Tognetti – alessandro.tognetti@unipi.it

Volume conductor theory

- We are targeting the study of electrical potentials generated by excitable tissues (neural, cardiac, muscular) and the stimulation of bio-tissues by means of external currents (e.g. neural stimulation, cardiac pacing)
- The theoretical framework is the **volume conductor theory** which describes the **conservation of current in a 3D conductive medium** (i.e., the volume conductor)
- Considering low frequency range the medium can be considered as purely resistive

Volume conductor theory



- This general sketch describes generation of biopotentials (ECG, EMG, EEG) and the stimulation of biological tissues (cuff nerve stimulation, pacing, defibrillation, FES)

Examples

A Bidomain Model of Epiretinal Stimulation

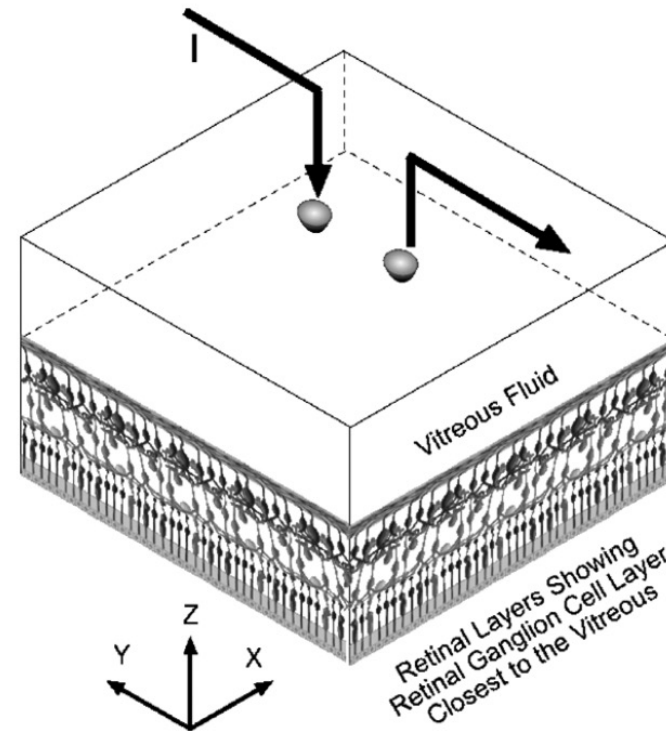
Socrates Dokos, *Member, IEEE*, Gregg J. Suaning, *Member, IEEE*, and Nigel H. Lovell, *Senior Member, IEEE*

Abstract—A computational bidomain model of epiretinal stimulation is presented, consisting of a continuum description of active retinal tissue in contact with bulk vitreous fluid. Results from two-electrode and four-electrode bipolar stimulation suggest that a biphasic cathodic-anodic stimulus sequence is effective in providing targeted focal activation of retinal tissue. Undesired secondary activations beneath each electrode return may be eliminated by using multiple returns for each stimulus electrode.

Index Terms—Bioelectric potentials, charge injection, electrodes, modeling, neuroprosthesis, visual system.

I. INTRODUCTION

DEVELOPMENT of an intraocular vision prosthesis for the blind requires that it be capable of delivering large numbers of parallel electrical stimuli to surviving retinal neurons in order to facilitate the generation of useful visual percepts. Such stimulation can be achieved via a regular array of bipolar stimulus current sources comprised of multiple current injection and return electrodes, not necessarily equal in number. The ensuing



Examples

A MULTI-SCALE SIMULATION OF RETINAL PHYSIOLOGY*

BELAL ABUELNASR[†] AND ADAM R. STINCHCOMBE[†]

Abstract. We present a detailed physiological model of the retina that includes the biochemistry and electrophysiology of phototransduction, neuronal electrical coupling, and the spherical geometry of the eye. The model is a parabolic-elliptic system of partial differential equations based on the mathematical framework of the bi-domain equations, which we have generalized to account for multiple cell-types. We discretize in space with non-uniform finite differences and step through time with a custom adaptive time-stepper that employs a backward differentiation formula and an inexact Newton method. A refinement study confirms the accuracy and efficiency of our numerical method. Numerical simulations using the model compare favorably with experimental findings, such as desensitization to light stimuli and calcium buffering in photoreceptors. Other numerical simulations suggest an interplay between photoreceptor gap junctions and inner segment, but not outer segment, calcium concentration. Applications of this model and simulation include analysis of retinal calcium imaging experiments, the design of electroretinograms, the design of visual prosthetics, and studies of ephaptic coupling within the retina.

Key words. Adaptive solver, inexact Newton method, Newton-iterative methods, retina model, simulation, biological model

MSC codes. 65N06, 35B27, 92-08, 92-10

1. Introduction. The retina is a unique part of the brain in that it is physically

Examples

scientific reports



OPEN

Simulating epileptic seizures using the bidomain model

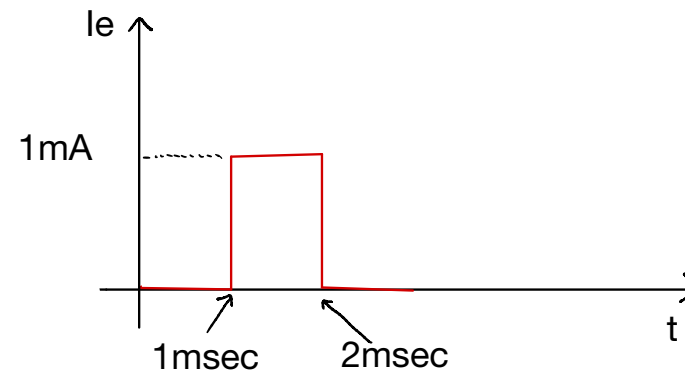
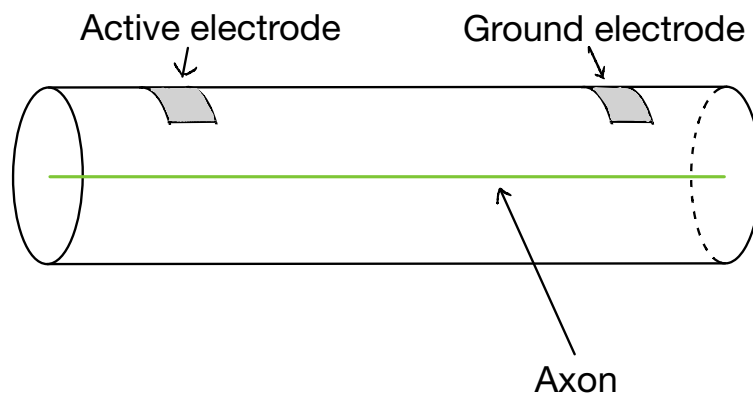
Jakob Schreiner^{1,2✉} & Kent-Andre Mardal^{1,2,3}

Epileptic seizures are due to excessive and synchronous neural activity. Extensive modelling of seizures has been done on the neuronal level, but it remains a challenge to scale these models up to whole brain models. Measurements of the brain's activity over several spatiotemporal scales follow a power-law distribution in terms of frequency. During normal brain activity, the power-law exponent is often found to be around 2 for frequencies between a few Hz and up to 150 Hz, but is higher during seizures and for higher frequencies. The Bidomain model has been used with success in modelling the electrical activity of the heart, but has been explored far less in the context of the brain. This study extends previous models of epileptic seizures on the neuronal level to the whole brain using the Bidomain model. Our approach is evaluated in terms of power-law distributions. The electric potentials were simulated in 7 idealized two-dimensional models and 3 three-dimensional patient-specific models derived from magnetic resonance images (MRI). Computed electric potentials were found to follow power-law distributions with slopes ranging from 2 to 5 for frequencies greater than 10–30 Hz.

Epilepsy is the fourth largest neurological disorder with a prevalence of 7.1 per thousand in all ages¹. The condition involves seizures of abnormal brain activity, where waves of electrical activity propagate through the brain and cause a variety of symptoms, ranging from absent-mindedness to loss of consciousness. An epileptic seizure is due to excessive or synchronous neural activity in the brain^{2,3}. Extensive modeling has been performed on the neuronal level, but only few works^{4,5}, have to the authors' knowledge, scaled up the neuronal modelling to brain-

Example 4: Axonal stimulation using nerve cuff electrodes

- Consider a $L=10$ mm length nerve bundle of radius $R=1$ mm.
- The bundle is stimulated by two cuff electrodes represented as squares of side length 1 mm wrapped around the bundle and having a center spacing of 7 mm along the main axis of the bundle
- The active left cuff electrode delivers an inward pulse current of 1mA over 1msec, the right electrode is held at ground. Considering an axon in the center of the bundle (aligned with the x-axis), derive:
 - The PDE governing intra- and extracellular potentials on the axon
 - Solve in FE for V_m
 - Plot the membrane potential over axon length at different times (1msec, 3msec, 5msec, 7msec, 9msec)
 - Plot the external potential in the point $(0.75*L, 0.25*R, 0)$



Volume conductor theory: methods

- The models we obtain are described by PDEs, which are equations that relate the partial derivatives of a function (i.e., the dependent variable) of several variables.
- Dependent variable: electric potential $\phi(\mathbf{x}, t)$
- Current sources: bioelectrical sources (membrane currents of the excitable tissues), external currents (injected by electrodes)
- General PDE model for conserved quantities
 - For conserved quantities such as mass, charge, heat or energy, we can use it to derive a general PDE
 - This general PDE states that the rate of increase of a conserved quantity plus the rate of outward flow ($\nabla \cdot \Gamma$) is equal to the rate of production (f).
- Solution methods
 - Analytical, closed form -> **can be very complex!**
 - Numerical solutions
 - Finite differences
 - Finite elements

General PDE for conserved quantities: conservation law formulation

$$\frac{\partial u(\mathbf{x}, t)}{\partial t} + \nabla \cdot \mathbf{\Gamma}(\mathbf{x}, t) = f(\mathbf{x}, t)$$

Conserved scalar quantity could be mass, charge, heat, energy.....

$$\mathbf{x} = (x, y, z)^T$$

$u(\mathbf{x}, t)$ Density of the conserved scalar quantity

Amount of the conserved scalar quantity per unit of volume

$\mathbf{\Gamma}(\mathbf{x}, t)$ Flux density

$\mathbf{\Gamma} \cdot \hat{\mathbf{n}} \, dS$ is the amount of the scalar quantity flowing per unit time across the infinitesimal area dS perpendicular to the direction $\hat{\mathbf{n}}$

$f(\mathbf{x}, t)$ Bulk source

Volume conductor: let's think about using charge as conserved quantity

Rate of production / consumption of u (unit of u per unit time)

Volume conductor theory

- Dependent variable: electric potential:

$$\phi(\mathbf{x}, t) \quad [\text{Volt}]$$

- Conserving quantity: electrical charge
- Conservation of charge law: total amount of charge remains fixed in a fixed volume

$$\frac{\partial u(\mathbf{x}, t)}{\partial t} = 0$$

- Flux term: current density

$$\mathbf{\Gamma} = \mathbf{J}(\mathbf{x}, t)$$

$\mathbf{J} \cdot \hat{\mathbf{n}} \, dS$ current (charge per unit time) flowing across the small area dS perpendicular to the direction $\hat{\mathbf{n}}$; Unit of current per area $[\text{A/m}^2]$

Volume conductor theory

- Bulk source: current source density

$$f = I_v(\mathbf{x}, t)$$

Production ($I_v > 0$) of electrical charge per unit time per unit volume (current per unit volume [A/m³])

- Ohm Law:

$$\mathbf{J} = \sigma \mathbf{E} \quad \sigma(\mathbf{x})$$

Inhomogeneous medium (conductivity depends on position); it could also be non isotropic

$$\nabla \cdot \mathbf{J}(\mathbf{x}, t) = I_v(\mathbf{x}, t) \quad \mathbf{E} = -\nabla \phi$$

$$\nabla \cdot (\sigma(\mathbf{x}) \nabla \phi(\mathbf{x}, t)) = -I_v(\mathbf{x}, t)$$

Volume conductor general PDE

Volume conductor theory

$$\nabla \cdot (\sigma(\mathbf{x}) \nabla \phi(\mathbf{x}, t)) = -I_v(\mathbf{x}, t)$$

homogeneous
medium

Active tissues with
bioelectric sources
(nerve, muscle, heart)

Passive tissues (no
bulk source)

$$\nabla^2 \phi(\mathbf{x}, t) = -\frac{I_v(\mathbf{x}, t)}{\sigma}$$

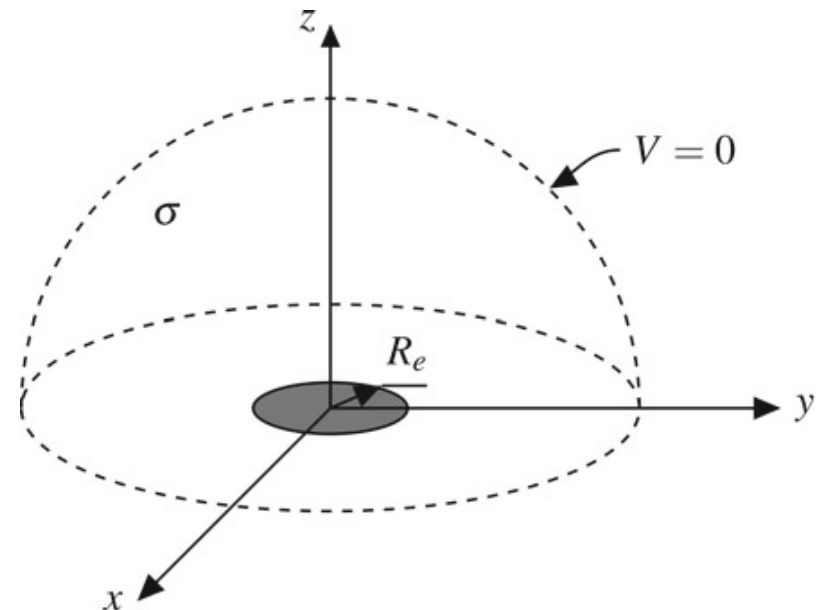
Poisson equation

$$\nabla^2 \phi(\mathbf{x}, t) = 0$$

Laplace equation

Example#1: Access Resistance of Electrode Disc

- 2D circular disc electrode embedded in a 3D hemispherical infinite medium of conductivity $\sigma = 1 \text{ S m}^{-1}$
- The radius of the disc electrode is $R_e = 5 \text{ mm}$
- The potential at the infinite boundary of the hemisphere is taken to be 0 .
- Outside the electrode, the x-y plane boundary surrounding the electrode is assumed to be electrically-insulating.
- The surface of the disc electrode is isopotential ($V_s = 1\text{V}$)
- Derive and plot the normal component of the current density on the electrode as a function of radial position and determine the total electrode current I_s
- Determine the access resistance of the electrode defined V_s/I_s
- Use of both analytical and numerical (FE) methods and make comparison about the current density on the electrode and the access resistance



Example#1: Governing equations

- Governing PDE:

$$\nabla \cdot \sigma \nabla \phi = 0 \longrightarrow \nabla^2 \phi = 0$$

Conductivity constant on the domain

- Current density:

$$\mathbf{J} = -\sigma \nabla \phi$$

- Access resistance:

$$Z = \frac{V_s}{I_s}$$

Electrode potential divided by the total current outflowing from the electrode

Example#1: boundary conditions

- On the electrode: $\longrightarrow \begin{cases} z = 0 \\ r \leq R_e \end{cases}$

$$\phi(r, z) = V_s$$

- xy plane outside the electrode: $\longrightarrow \begin{cases} z = 0 \\ r > R_e \end{cases}$

$$J_z(r, z) = 0 \implies \frac{\partial \phi}{\partial z} = 0$$

Example#1: Analytical solution

- Voltage is axisymmetric about the z-axis
- Method of separation of variables
 - It is possible to obtain the exact closed-form solution by using the method of separation of variables. The method assumes that the dependent variable of the PDE can be split into a product of separate functions of the independent variables

$$\phi(r, z) = P(r) Q(z)$$

$$r = \sqrt{x^2 + y^2}$$

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

Example#1: Analytical solution

- Methods: symbolic calculations using matlab live editor (“electro_disc.mlx”)
 - Matlab live editor: possibility to view results together with the code that produced them. Equations, images, hyperlinks, and formatted text can be mixed with the code to document.
 - Support for the symbolic calculation is crucial
 - Note: the analytical solution is optional in the 2025/26 program!

Example#1: Separation of variables

Note: analytical solution is optional!

$$\frac{\frac{\partial}{\partial r} P(r) + \frac{\partial^2}{\partial r^2} P(r)}{P(r)} = -\frac{\frac{\partial^2}{\partial z^2} Q(z)}{Q(z)}$$
$$\frac{\frac{\partial}{\partial r} P(r) + \frac{\partial^2}{\partial r^2} P(r)}{P(r)} = \lambda$$
$$\frac{\frac{\partial^2}{\partial z^2} Q(z)}{Q(z)} = -\lambda$$

Each side of the expression is a function of r or z alone, then each term must be equal to a constant (λ)

We obtain two ordinary differential equations (ODEs) that can be solved analytically

Example#1: Analytical solution

- Results

On the electrode $\left[\begin{array}{l} z = 0 \\ r < R_e \end{array} \right. \longrightarrow J_z = \frac{2 V_s \sigma}{\pi \sqrt{R_e^2 - r^2}}$

$$I_s = \int_0^{R_e} J_z 2\pi r dr = 4 R_e V_s \sigma = 20 \text{ mA}$$

$$R_s = \frac{1}{4 R_e \sigma} = 50 \text{ } \Omega$$

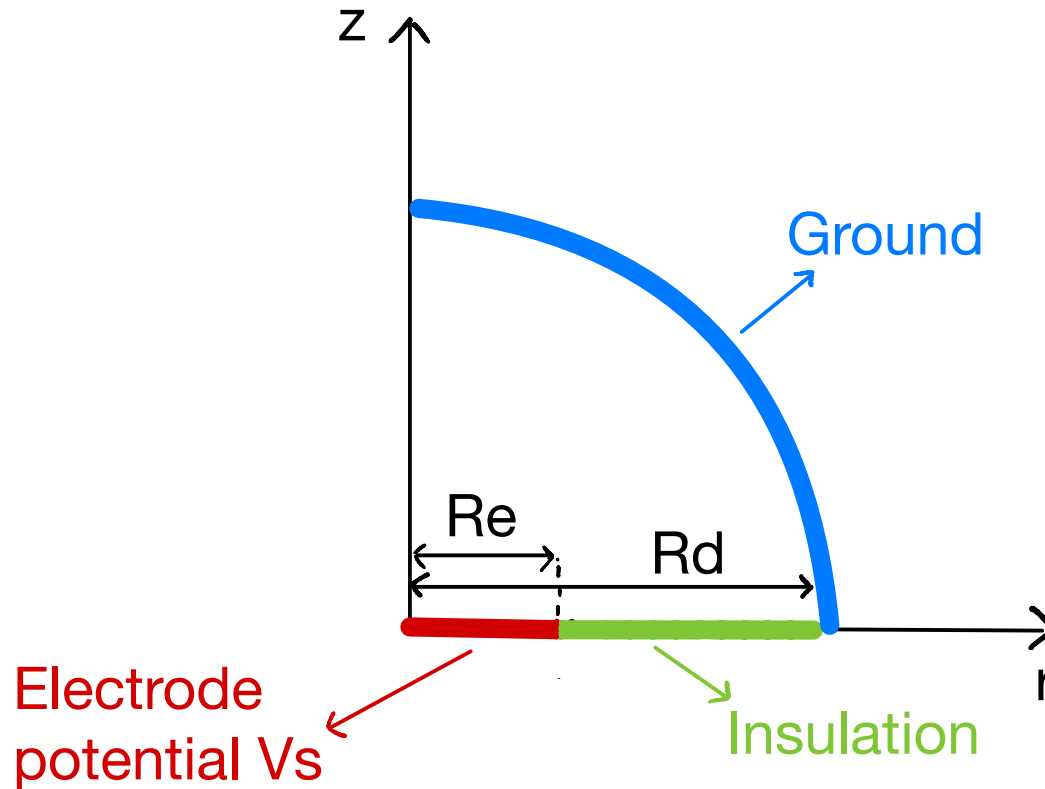
Example#1: Analytical solution

- Comments
 - The example PDE is relatively simple, however to find the closed form solution is very complex
 - For most PDE systems of our interest, the closed form solution does not exist!
 - Most PDE systems can be more easily solved with numerical methods such as finite differences and finite elements

Think about how to solve the problem of electrode disc with FE using Comsol !

Example#1: Finite element solution in COMSOL

- Axisymmetric problem about z-axis
 - “electrode_disc_FE.mph”



Note: $R_d \gg R_e$ to simulate ground for $r=\infty$

Example#1: Finite element solution in COMSOL

- Model Wizard
 1. Select 2D axisymmetric
 2. Select Physics: AC/DC | Electric currents (use the default name of the dependent variable V)
 3. Study: select stationary
- Global definition
 - Right-click the Global definition subnode and select parameters ($V_s=1$ [V], $\sigma=1$ [S/m], $R_e=5$ [mm], $Z_{th}=1/(4*\sigma*R_e)$, $R_d=50$ mm)
 - Z_{th} : theoretical access resistance (obtained with the analytical solution)

Example#1: Finite element solution in COMSOL

- Geometry
 1. Select the Geometry1 node. In the settings window, change the default length unit to 'mm'.
 2. Right-click Geometry 1 and select Circle. Specify the radius as R_d and the sector angle as 90° . Click Build Selected .
 3. To draw the electrode as a line interval of length R_e (5 mm), right-click geometry 1 and select Polygon. In the settings window, specify the r coordinate values as (0, R_e) and the z coordinate (0, 0). Click Build Selected

Example#1: Finite element solution in COMSOL

- Component definitions
 1. Right-click the Definitions sub-node under the Component 1 node, and select Non Local Couplings | Integration. Specify the geometric entity level as “boundary”, and select boundary 2 corresponding to the electrode. This integration operator will return the integral of its argument over the electrode and we will use it to determine the total electrode current I_s .
 2. Right-click the Definitions sub-node under the Component 1 node, and select “Variables”. Keep the default geometric entity selection as “Entire model”. Define I_s with expression $\text{intop1}(\text{ec.nJ})$. This will return the total electrode current. Specify the theoretical current density as $2 \cdot \sigma \cdot V_s / (\pi \cdot \sqrt{R_e^2 - r^2})$. Specify Z_s as V_s / I_s

$$I_s = \int_0^{R_e} J_z 2\pi r dr$$

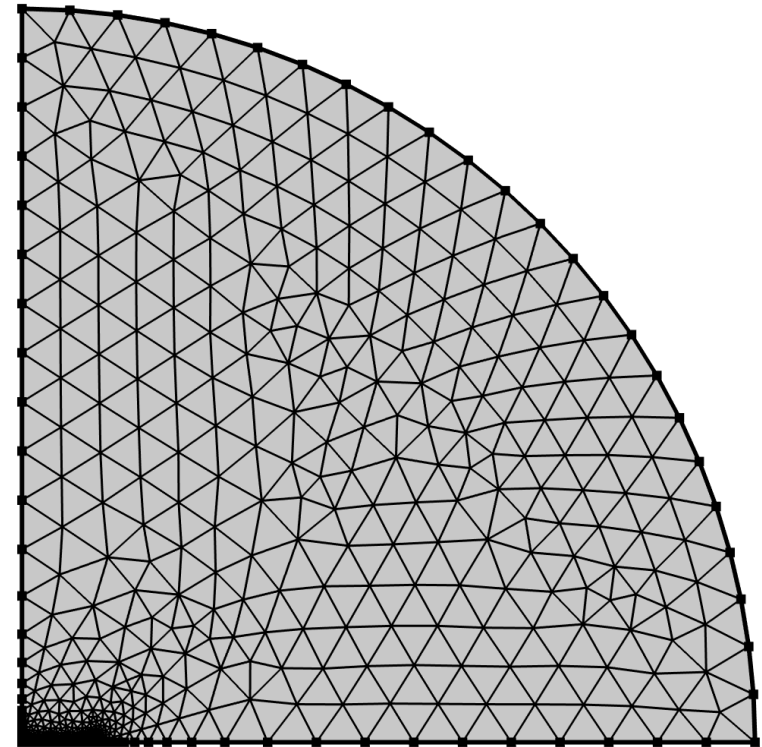
$$J_z = \frac{2 V_s \sigma}{\pi \sqrt{R_e^2 - r^2}}$$

Example#1: Finite element solution in COMSOL

- Electric currents
 1. Select the Current Conservation 1 sub-node of Electric Currents. In the settings panel, specify the electrical conductivity as 'user-defined' with a value of σ . Similarly, specify the relative permeability as user-defined, leaving the default value of 1.
 2. Right-click the Current Conservation 1 sub-node and select 'Electric Potential'. In the settings panel, select boundary 2 (i.e. electrode disc) and specify the electric potential as V_s .
 3. Right-click the Current Conservation 1 sub-node again and select 'Ground'. In the settings panel, select boundary 4 to be the ground ($r=R_d$).

Example#1: Finite element solution in COMSOL

- Mesh
 1. Right-click the Mesh1 node and select “Distribution”. In the settings panel, select boundary 2 and specify 100 as the number of elements.
 2. Right-click the Mesh 1 node again and select “Free Triangular” leaving the geometric entity level to the default value of “Remaining”. Click the Build all button to build and display the mesh.

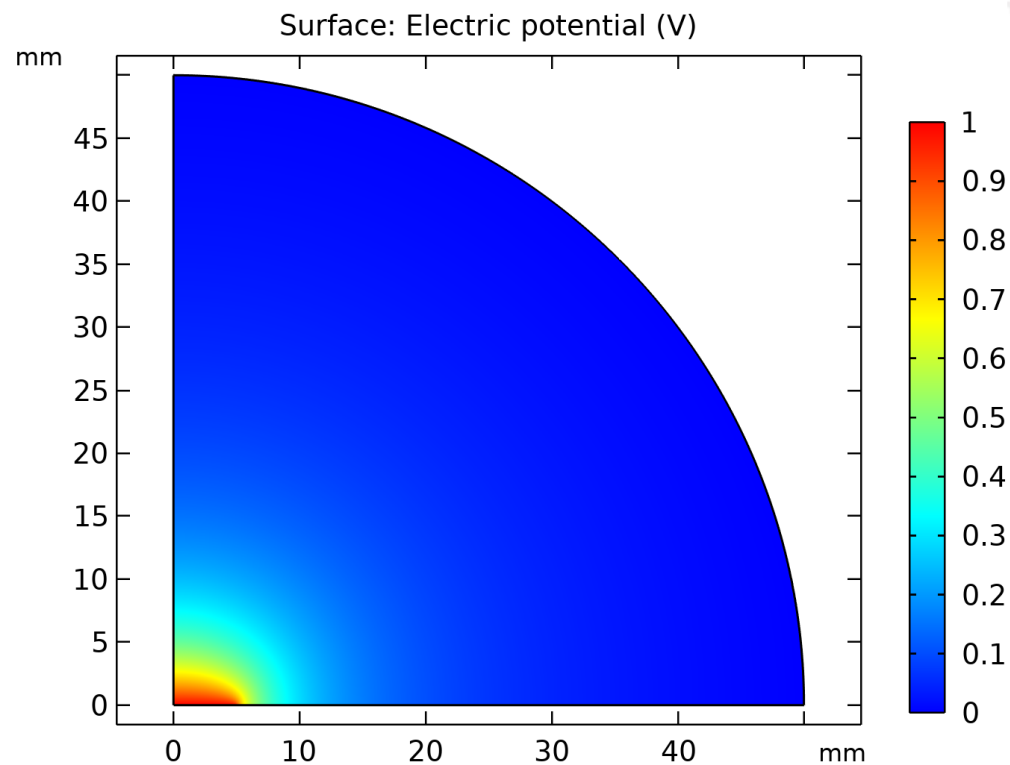


Mesh for axisymmetric disc electrode geometry. The higher density of elements can be seen over the electrode disc on the lower left boundary

Example#1: Finite element solution in COMSOL

- Study

1. Right-click study 1 and select compute. This will compute the solution and display the default plot of voltage



Computed voltage distribution around disc electrode. The electrode at *bottom left* is at an iso-potential level of 1 V, whilst the boundary at Rd is at 0 V

Example#1: Finite element solution in COMSOL - Results

1. Plot the computed J_z vs. r on the disc electrode: right-click the Results node and select “1DPlotGroup”. Right-click the 1D Plot Group 3 sub-node and select ‘Line Graph’. In the Settings, select boundary 2. Under the y-Axis Data tab, specify $ec.nJ$ as the expression to plot. For the x-Axis Data under Parameter, select “Expression” and specify r . Under the Legends tab, check the “Show leg-ends” checkbox and specify the “Manual” option. In the Legends table, enter the text “Computed”.
2. Plot the theoretical value of J_z vs r on the disc electrode: right-click the 1D Plot Group 3 sub-node and select “Line Graph”. In the Settings, select boundary 2 and specify J_{th} as the expression to plot. For the x-Axis Data under Parameter, select “Expression” and specify r . Under the Legends tab, check the “Show legends” checkbox and specify the “Manual” option. In the Legends table, enter the text “Theoretical”. Click the Plot button to display the current density plots. As can be seen from the plots, the computed and theoretical solutions for current density across the electrode agree.
3. Right-click the Derived Values sub-node under Results and select “Global Evaluation”. In the settings panel, specify Z_s as the expression to evaluate. Click the Evaluate button to display the value of the access resistance. COMSOL determines this value to be $49, \dots \Omega$ which is slightly lower than the 50Ω theoretical value. The discrepancy is due to the mesh resolution not being able to capture the theoretical infinite current density at the edge of the disc.

Example#1: Simple example, can be extended to complex studies



Heart Rhythm

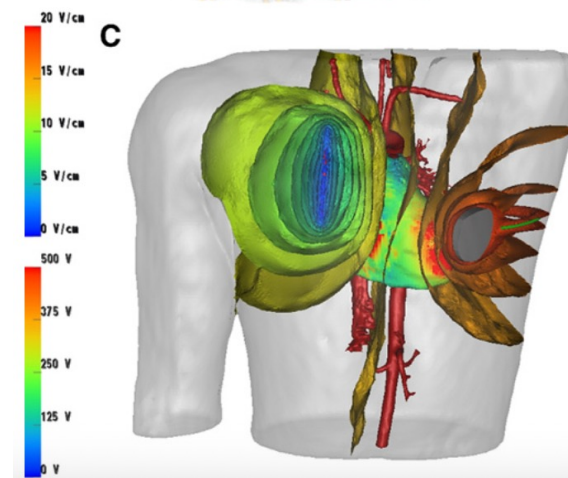
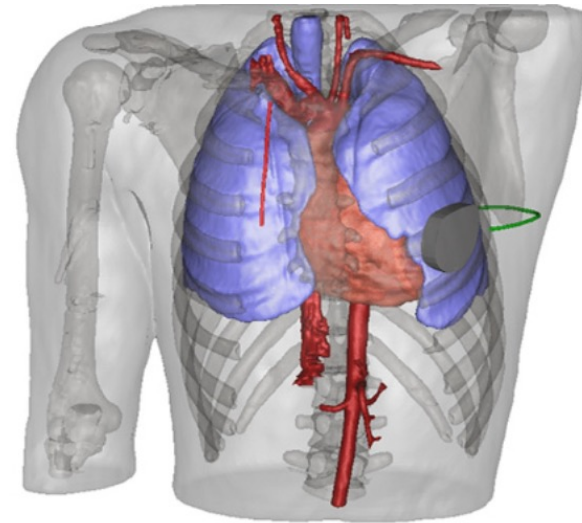
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Experimental

Finite element modeling of subcutaneous implantable defibrillator electrodes in an adult torso

Matthew Jolley MD ^{*}, Jeroen Stinstra PhD [†], Jess Tate BS [†], Steve Pieper PhD [‡], Rob MacLeod PhD [†], Larry Chu MD, MPH ^{*}, Paul Wang MD, FHRS [§], John K. Triedman MD, FHRS [¶]  

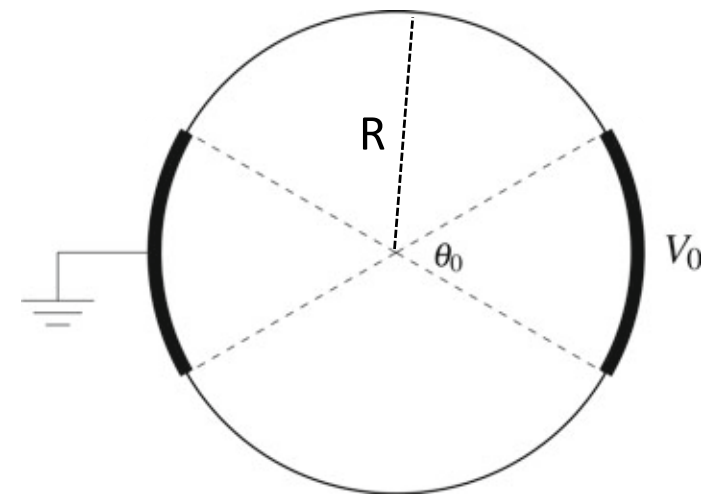


Example#2: Effect of electric field on a Petri Dish

- A Petri dish is a shallow transparent lidded circular dish that biologists use for cell culture.
- To determine the effect of electric fields on cells in the Petri Dish the electric field is delivered using an identical pair of wire electrodes. formed from circular arcs placed at opposite ends of the dish against its walls. Electrodes subtend the angle θ_0 at the center of the dish.
- One electrode is grounded, the other has V_0 potential.
- All the other boundaries are electrically insulating.
- The conductivity of the saline medium in which the cells are cultured is $\sigma = 1 \text{ S m}^{-1}$ (the medium is homogeneous)
- The dish is filled to a height of 10 mm (useful for FE implementation)

Find:

1. The PDE governing voltage distribution in polar coordinates (r, θ) ; $r=0$ is the center of the dish; $\theta=0$ is the x-axis (electric potential $\phi(\mathbf{x})$ with $\mathbf{x}=[r \ \theta]^T$).
2. Solve the PDE using the method of finite differences with: $R=45 \text{ mm}$, $\theta_0=60^\circ$, $V_0=2\text{V}$.
3. Given R and V_0 , find the optimal value of θ_0 that yields the largest area in which $|\mathbf{E}|$ is uniform within $\pm 10\%$ of the value in the center of the dish
4. Solve the PDE using finite elements and compare



Example#2: Governing equations

- Governing PDE:

$$\nabla \cdot \sigma \nabla \phi = 0 \longrightarrow \nabla^2 \phi = 0$$

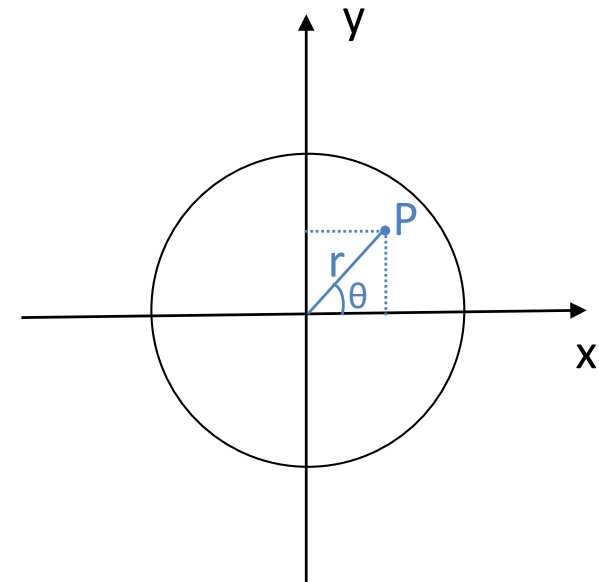
Conductivity constant on the domain

- Bidimensional problem:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

- Polar coordinates:

$$r = \sqrt{x^2 + y^2} \quad x = r \cos(\theta) \quad y = r \sin(\theta)$$



Example#2: Governing equations

- Governing PDE in polar coordinates
 - Symbolic calculations in Matlab live editor (“PetriDish_PDE_symbolic.mlx”) – optional in 2025/26 program

$$\frac{1}{r} \frac{\partial \phi(r, \theta)}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \phi(r, \theta)}{\partial \theta^2} + \frac{\partial^2 \phi(r, \theta)}{\partial r^2} = 0$$

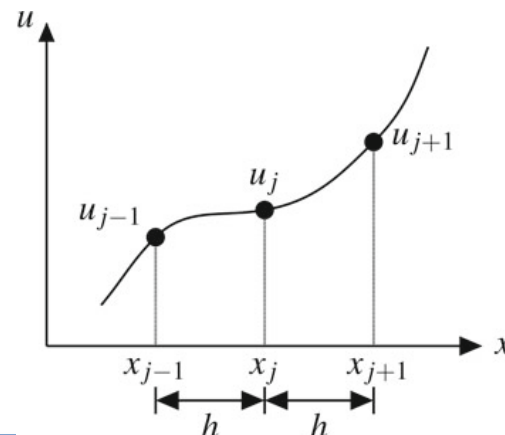
- We will solve the above PDE by using the finite difference method

Finite difference method

- Simple approach to numerically solve PDEs
- The PDE is discretized into a regular grid covering the PDE domain
- The partial derivatives are approximated from the discretized grid variables
- For example, we can evaluate the dependent variable u on a discretized 1D spatial domain made of a set of regularly-spaced (h) nodes each spaced a distance h apart

We denote the value of u at node j as u_j

$x_1, x_2, \dots, x_j, \dots$
 $\downarrow \quad \downarrow \quad \downarrow$
 $u_1, u_2, \dots, u_j, \dots$



Finite difference method

- We can use Taylor's theorem on the grid variables to obtain the approximation of the first and second derivatives of the dependent variable
- **Forward finite difference approximation to the first derivative at x_j**

$$\left. \frac{\partial u}{\partial x} \right|_{x=x_j} \approx \frac{u_{j+1} - u_j}{h}$$

First order Taylor's
polynomial

$$\longrightarrow u_{j+1} = u_j + h \left. \frac{\partial u}{\partial x} \right|_{x=x_j} + \mathcal{O}(h^2)$$

$$\downarrow$$
$$\left. \frac{\partial u}{\partial x} \right|_{x=x_j} = \frac{u_{j+1} - u_j}{h} + \mathcal{O}(h)$$

Finite difference method

- Backward finite difference approximation to the first derivative at x_j

$$\left. \frac{\partial u}{\partial x} \right|_{x=x_j} \approx \frac{u_j - u_{j-1}}{h}$$

First order Taylor's
polynomial

$$\longrightarrow u_{j-1} = u_j - h \left. \frac{\partial u}{\partial x} \right|_{x=x_j} + \mathcal{O}(h^2)$$



$$\left. \frac{\partial u}{\partial x} \right|_{x=x_j} = \frac{u_j - u_{j-1}}{h} + \mathcal{O}(h)$$

Finite difference method

- Finite difference approximation to the second derivative at x_j

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{x=x_j} \approx \frac{u_{j+1} - 2u_j + u_{j-1}}{h^2}$$

$$u_{j+1} = u_j + h \left. \frac{\partial u}{\partial x} \right|_{x=x_j} + \frac{h^2}{2} \left. \frac{\partial^2 u}{\partial x^2} \right|_{x=x_j} + \frac{h^3}{6} \left. \frac{\partial^3 u}{\partial x^3} \right|_{x=x_j} + \mathcal{O}(h^4)$$

$$u_{j-1} = u_j - h \left. \frac{\partial u}{\partial x} \right|_{x=x_j} + \frac{h^2}{2} \left. \frac{\partial^2 u}{\partial x^2} \right|_{x=x_j} - \frac{h^3}{6} \left. \frac{\partial^3 u}{\partial x^3} \right|_{x=x_j} + \mathcal{O}(h^4)$$

Third order Taylor's
polynomials

Sum

$$u_{j+1} + u_{j-1} = 2u_j + h^2 \left. \frac{\partial^2 u}{\partial x^2} \right|_{x=x_j} + \mathcal{O}(h^4)$$

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{x=x_j} = \frac{u_{j+1} - 2u_j + u_{j-1}}{h^2} + \mathcal{O}(h^2)$$

Finite difference method

- FD approximations to derivatives can be substituted into a specific PDE to obtain a system of equations in the nodal variables u_j
 - How many equations? ?
- The result is the quantification of the dependent variable on the discretized domain

Example#2: finite difference implementation

- Regular grid over $\mathbf{x} = (r, \theta)^T$
- Symmetry of the problem about the x-axis, we will use only half domain $r \in [0, R]$ $\theta \in [0, \pi]$
- Grid definition:

$$r_i = (i - 1) \Delta r \quad i = 1 \dots N + 1$$
$$\theta_j = (j - 1) \Delta \theta \quad j = 1 \dots M + 1$$

Diagram illustrating the grid boundaries with blue arrows:

- $r_1 = 0$ (pointing to the first grid point)
- $r_{N+1} = R$ (pointing to the last grid point)
- $\theta_1 = 0$ (pointing to the first grid point)
- $\theta_{N+1} = \pi$ (pointing to the last grid point)

$$\Delta r = \frac{R}{N} \quad \Delta \theta = \frac{\pi}{M}$$

Example#2: finite difference implementation

- Discretized PDE

$$\frac{1}{r_i} \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta r} + \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2} + \frac{1}{r_i^2} \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta \theta^2} = 0$$

- The discretized PDE is undefined for $r_i=0$ so it is valid for $i>1$. We cannot use this approximation at the center of the domain. This is due to the choice of polar coordinates and not to a real discontinuity in the problem. We will approximate the potential at the center of the domain by considering the mean value of the surrounding nodes.

$$\phi_{1,j} = \frac{1}{M+1} \sum_{j=1}^{M+1} \phi_{2,j}$$

Example#2: finite difference implementation

- Discretized PDE

$$\frac{1}{r_i} \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta r} + \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2} + \frac{1}{r_i^2} \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta \theta^2} = 0$$

$\leftarrow r_i = (i-1) \Delta r$
 $\leftarrow \theta_j = (j-1) \Delta \theta$

$$\frac{1}{(i-1) \Delta r} \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta r} + \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2} + \frac{1}{(i-1)^2 \Delta r^2} \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta \theta^2} = 0$$

Multiplying by $\longrightarrow (i-1)^2 \Delta r^2 \Delta \theta^2$

$$(\phi_{i+1,j} - \phi_{i,j})(i-1) \Delta \theta^2 + (\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j})(i-1)^2 \Delta \theta^2 + (\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}) = 0$$

Collection of variables

$$((i-1)^2 \Delta \theta^2 + (i-1) \Delta \theta^2) \phi_{i+1,j} + ((i-1)^2 \Delta \theta^2) \phi_{i-1,j} - (2 + (i-1) \Delta \theta^2 + 2(i-1)^2 \Delta \theta^2) \phi_{i,j} + \phi_{i,j+1} + \phi_{i,j-1} = 0$$

Example#2: finite difference implementation

- Discretized PDE

$$k_1 \phi_{i+1,j} + k_2 \phi_{i-1,j} + k_3 \phi_{i,j} + \phi_{i,j+1} + \phi_{i,j-1} = 0$$

$$c_1 = (i-1)^2 \Delta\theta^2$$

$$c_2 = (i-1) \Delta\theta^2$$

$$k_1 = c_1 + c_2$$

$$k_2 = c_1$$

$$k_3 = -(2 + 2c_1 + c_2)$$

k_1, k_2, k_3
depends on i

- Boundary conditions on the dish walls $\rightarrow r_i = R \Rightarrow i = N + 1$

$$\phi_{N+1,j} = \begin{cases} \phi_{N+1,j} = V_0 & \theta_j = (j-1) \Delta\theta \leq \frac{\theta_0}{2} \\ \phi_{N+1,j} = 0 & \theta_j = (j-1) \Delta\theta \geq \pi - \frac{\theta_0}{2} \\ \phi_{N+1,j} - \phi_{N,j} = 0 & \frac{\theta_0}{2} < \theta_j = (j-1) \Delta\theta < \pi - \frac{\theta_0}{2} \end{cases}$$

active electrode

ground electrode

zero flux

insulation

Example#2: finite difference implementation

- Zero flux on the dish wall outside the electrodes
($r_i=R$ and $\theta_0/2 < \theta_j < \pi - \theta_0/2$)

$$\mathbf{J} \cdot \hat{\mathbf{n}} = 0 \quad \hat{\mathbf{n}} = \hat{\mathbf{r}} \quad \mathbf{J} = -\sigma \nabla \phi \Rightarrow \nabla \phi \cdot \hat{\mathbf{r}} = \frac{\partial \phi}{\partial r} = 0 \Rightarrow \phi_{N+1,j} = \phi_{N,j}$$

- Symmetry about the x-axis
 - It is equivalent to a zero-flux boundary condition for ϕ on the lower edge of the half-domain, corresponding the x-axis
 - X-axis: polar coordinates $\rightarrow j = 1$ ($\theta=0$) and $j=M+1$ ($\theta=\pi$).

Example#2: finite difference implementation

- Symmetry about the x-axis

$$\phi(x, y) = \phi(x, -y) \Rightarrow \phi(x, \Delta y) = \phi(x, -\Delta y)$$

$$\frac{\partial \phi}{\partial y} \Big|_{y=0} \approx \frac{\phi(x, \Delta y) - \phi(x, -\Delta y)}{2 \Delta y} \rightarrow \text{zero for the symmetry}$$

$$2 \frac{\partial \phi}{\partial y} \Big|_{y=0} \approx \frac{\phi(x, \Delta y) - \phi(x, 0)}{\Delta y} + \frac{\phi(x, 0) - \phi(x, -\Delta y)}{\Delta y} = \frac{\phi(x, \Delta y) - \phi(x, -\Delta y)}{\Delta y}$$

See Matlab calculations

$$\frac{\partial \phi}{\partial y} \Big|_{y=0} = 0 \Rightarrow \left[\frac{y}{r} \frac{\partial \phi}{\partial r} + \frac{x}{r^2} \frac{\partial \phi}{\partial \theta} \right]_{y=0} = \left[\frac{x}{r^2} \frac{\partial \phi}{\partial \theta} \right]_{y=0} \Rightarrow \frac{\partial \phi}{\partial \theta} \Big|_{y=0} = 0$$

$$\frac{\partial \phi}{\partial \theta} \Big|_{y=0} = 0 \xrightarrow{\text{FD approximation}} \left. \begin{aligned} \phi_{i,1} &= \phi_{i,2} \\ \phi_{i,M+1} &= \phi_{i,M} \end{aligned} \right\} \text{Conditions to set}$$

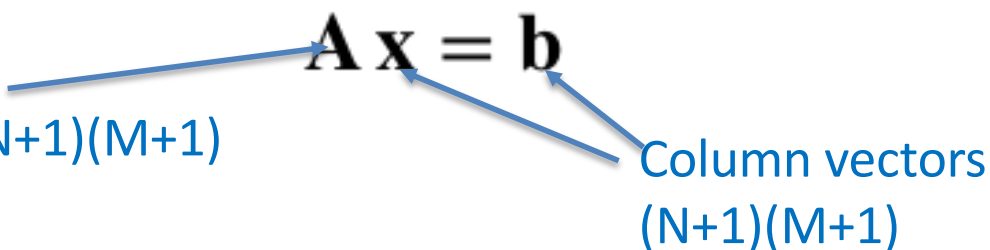
Example#2: finite difference implementation

- Each node contributes to an algebraic equation of the dependent variables evaluated on the grid
 - We want to calculate the electric potential in each point of the grid
 - We obtain $(N+1)*(M+1)$ equations
 - The unknowns are the grid variables
- Can be solved in matrix form

Square matrix
 $(N+1)(M+1) \times (N+1)(M+1)$

$\mathbf{A} \mathbf{x} = \mathbf{b}$

Column vectors
 $(N+1)(M+1)$



Example#2: finite difference implementation

- $\mathbf{x}$ vector of unknowns

$$\mathbf{x} = [\underbrace{\phi_{1,1} \cdots \phi_{1,M+1}}_{M+1 \text{ terms}} \underbrace{\phi_{2,1} \cdots \phi_{2,M+1}}_{M+1 \text{ terms}} \cdots \underbrace{\phi_{N+1,1} \cdots \phi_{N+1,M+1}}_{M+1 \text{ terms}}]^T$$

N+1 blocks

- h index of $\phi_{i,j}$ in $\mathbf{x} \rightarrow h = (M + 1)(i - 1) + j$
- h increment for a unit step in $i \rightarrow i \pm 1 \Rightarrow h \pm (M + 1)$
- h increment for a unit step in $j \rightarrow j \pm 1 \Rightarrow h \pm 1$

Example#2: finite difference implementation

- Steps
 1. Define the grid ($N+1$ steps in i , $M+1$ step in j ; in our example: $N=100$, $M=100$)
 2. Scroll through all the indexes of the grid (for loop in Matlab)
 3. For each couple of i, j we obtain an equation that has to be expressed in matrix form (6 different cases: domain, 3 external boundaries, center of the dish ($r=0 \rightarrow i=1$) and x-axis ($j=1$ and $j=M+1$))
 4. i, j couple $\rightarrow$ h-st equation

Example#2: finite difference implementation

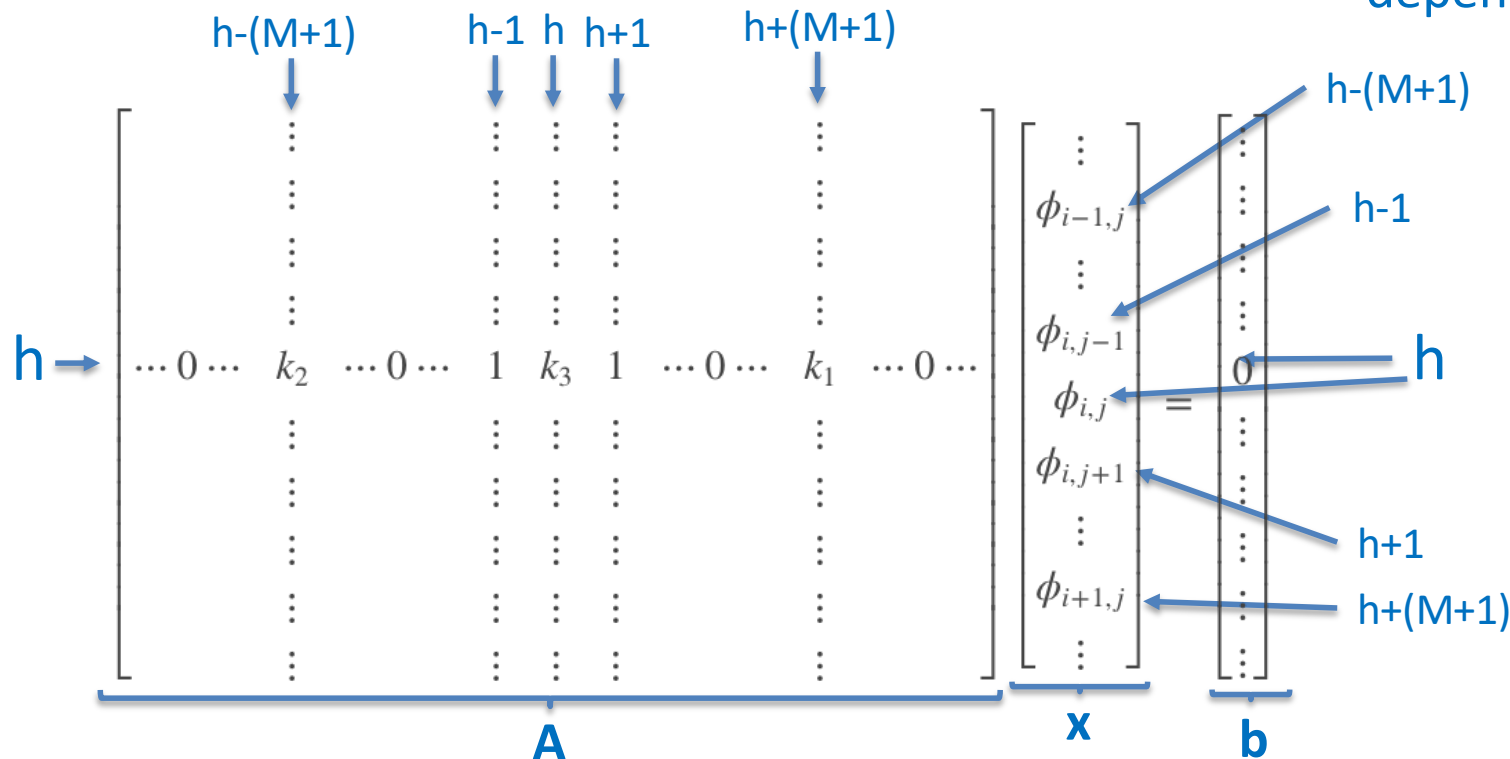
- Equation on the domain

```

else    %in the domain
    A(h,h)=k3;    %i,j -> h=(M+1)*(i-1)+j
    A(h,h-(M+1))=k2; %i-1,j ->h-(M+1)
    A(h,h+(M+1))=k1; %i+1,j -> h+(M+1)
    A(h,(h+1))=1; %i,j+1 -> h+1
    A(h,(h-1))=1; %i,j-1 -> h-1
end
    
```

$$k_1 \underbrace{\phi_{i+1,j}}_{h+(M+1)} + k_2 \underbrace{\phi_{i-1,j}}_{h-(M+1)} + k_3 \underbrace{\phi_{i,j}}_h + \underbrace{\phi_{i,j+1}}_{h+1} + \underbrace{\phi_{i,j-1}}_{h-1} = 0$$

Note: k1, k2, k3 depends on i

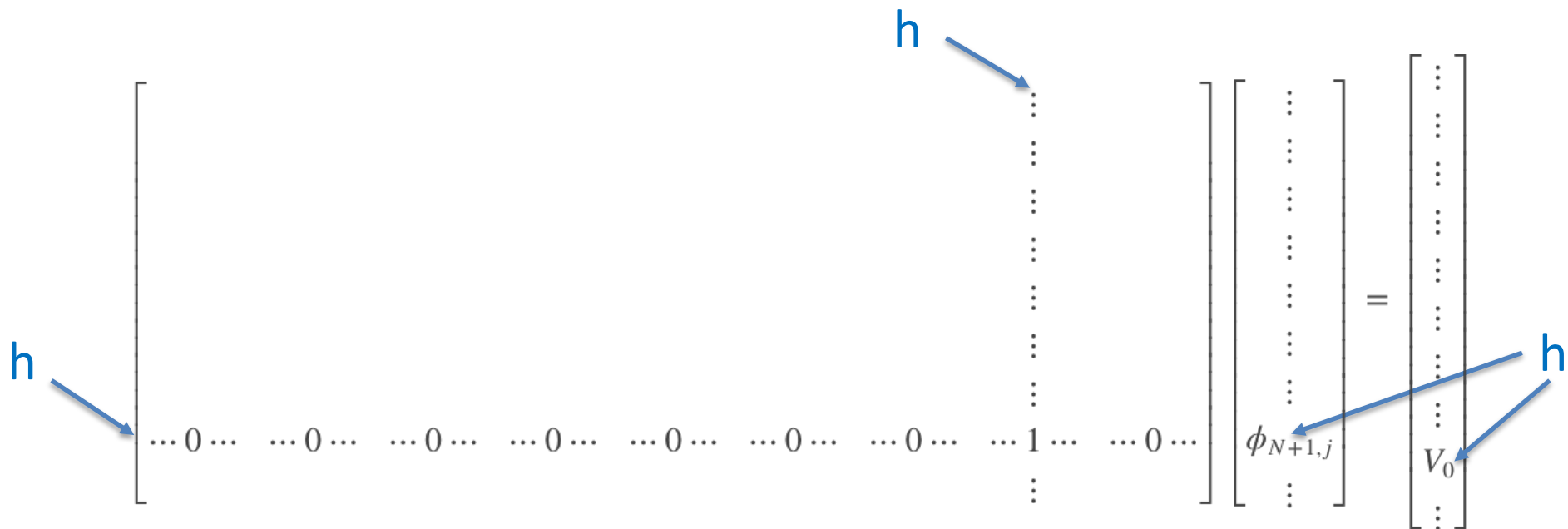


Example#2: finite difference implementation

- Equation on the active electrode (circular boundary)

$$\phi_{N+1,j} = V_0 \begin{cases} i = N + 1 \\ \theta_j \leq \frac{\theta_0}{2} \end{cases}$$

```
elseif (i==N+1) %circular boundary
    theta=(j-1)*d_theta;
    if (theta <= theta_0/2) %active electrode
        B(h)=V0;
        A(h,h)=1;
    elseif (theta >= pi-theta_0/2) %ground electrode
        A(h,h)=1; %B(index) is 0 by default
    else %electrical insulation
        A(h,h)=1;
        A(h,h-(M+1))=-1;
    end
```

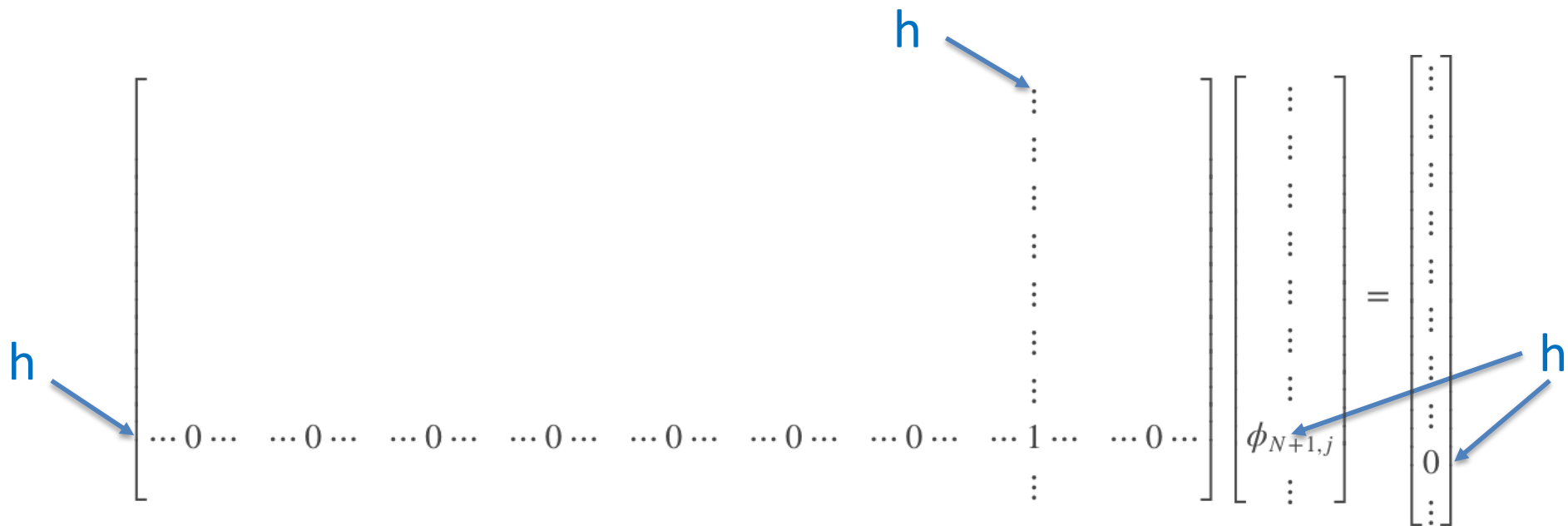


Example#2: finite difference implementation

- Equation on the ground electrode (circular boundary)

$$\phi_{N+1,j} = 0 \quad \begin{cases} i = N + 1 \\ \theta_j \geq \pi - \frac{\theta_0}{2} \end{cases}$$

```
elseif (i==N+1) %circular boundary
    theta=(j-1)*d_theta;
    if (theta <= theta_0/2) %active electrode
        B(h)=V0;
        A(h,h)=1;
    elseif (theta >= pi-theta_0/2) %ground electrode
        A(h,h)=1; %B(index) is 0 by default
    else %electrical insulation
        A(h,h)=1;
        A(h,h-(M+1))=-1;
    end
```

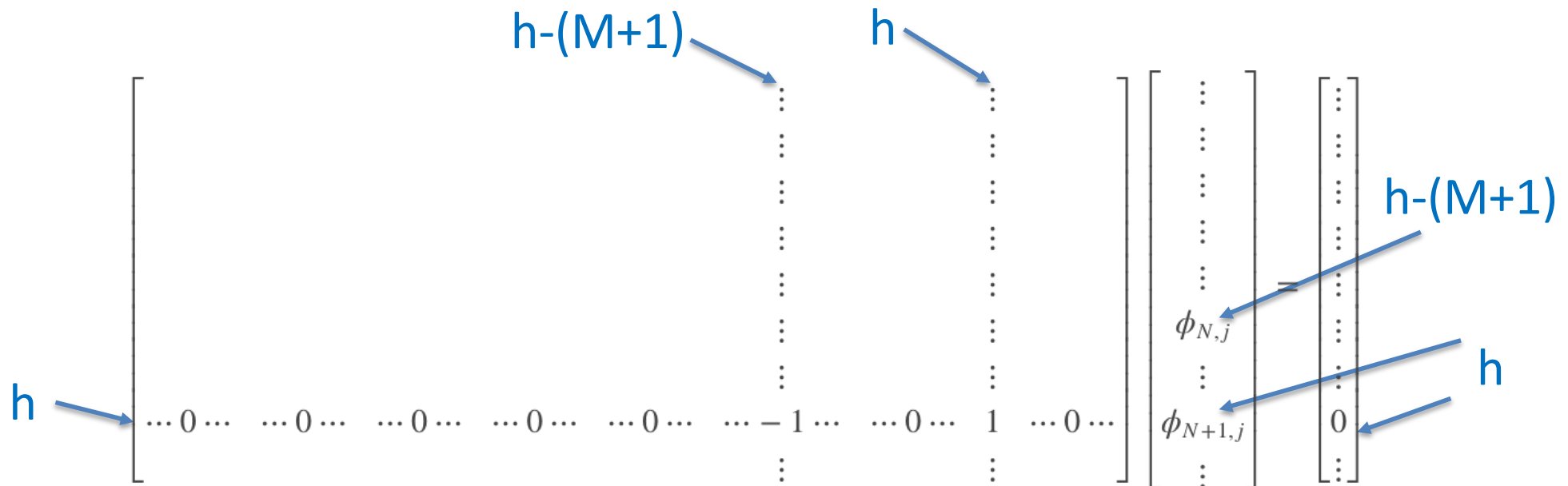


Example#2: finite difference implementation

- Equation insulating part of the circular boundary

$$\phi_{N+1,j} - \phi_{N,j} = 0 \begin{cases} i = N + 1 \\ \frac{\theta_0}{2} < \theta_j < \pi - \frac{\theta_0}{2} \end{cases}$$

```
elseif (i==N+1) %circular boundary
    theta=(j-1)*d_theta;
    if (theta <= theta_0/2) %active electrode
        B(h)=V0;
        A(h,h)=1;
    elseif (theta >= pi-theta_0/2) %ground electrode
        A(h,h)=1; %B(index) is 0 by default
    else %electrical insulation
        A(h,h)=1;
        A(h,h-(M+1))=-1;
    end
end
```

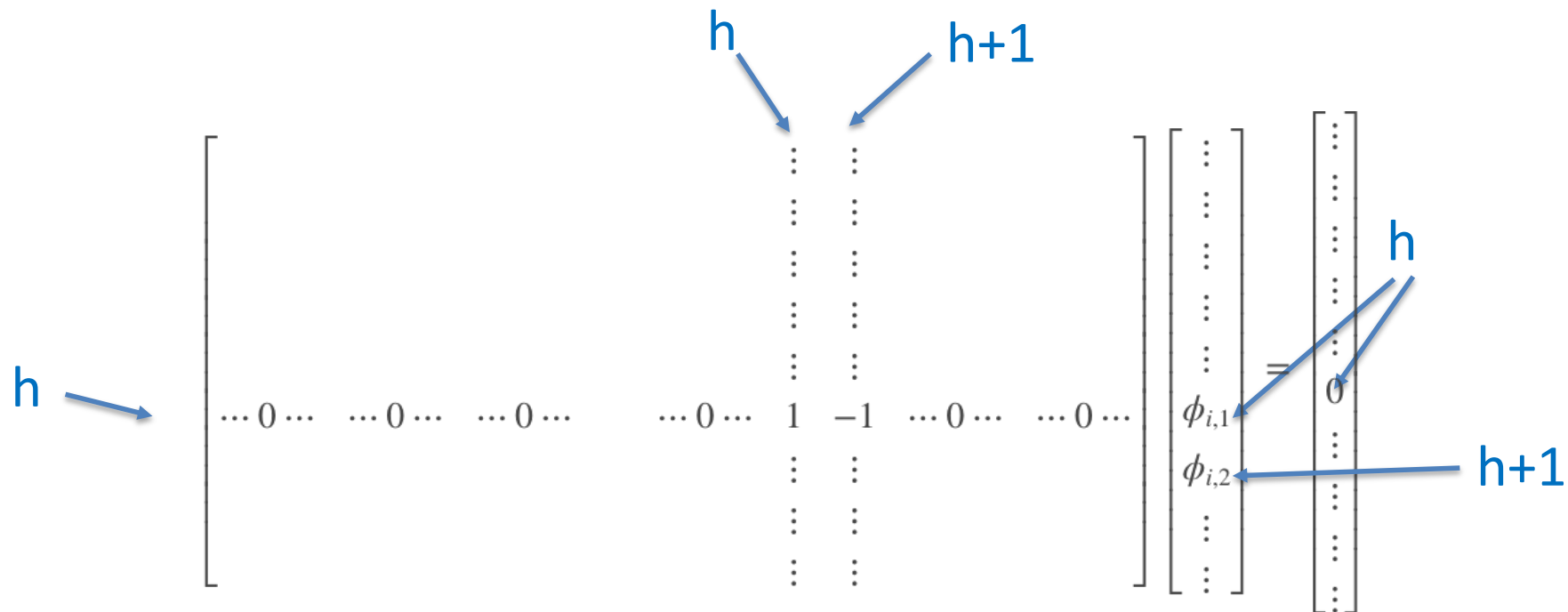


Example#2: finite difference implementation

- Zero flux on the x-axis

$$\begin{cases} \phi_{i,1} - \phi_{i,2} = 0 \\ j = 1 \end{cases}$$

```
elseif (j==1) %x-axis (thetaj=0)
    A(h,h)=1;
    A(h,h+1)=-1;
elseif (j==M+1) %x-axis (thetaj=pi)
    A(h,h)=1;
    A(h,h-1)=-1;
```

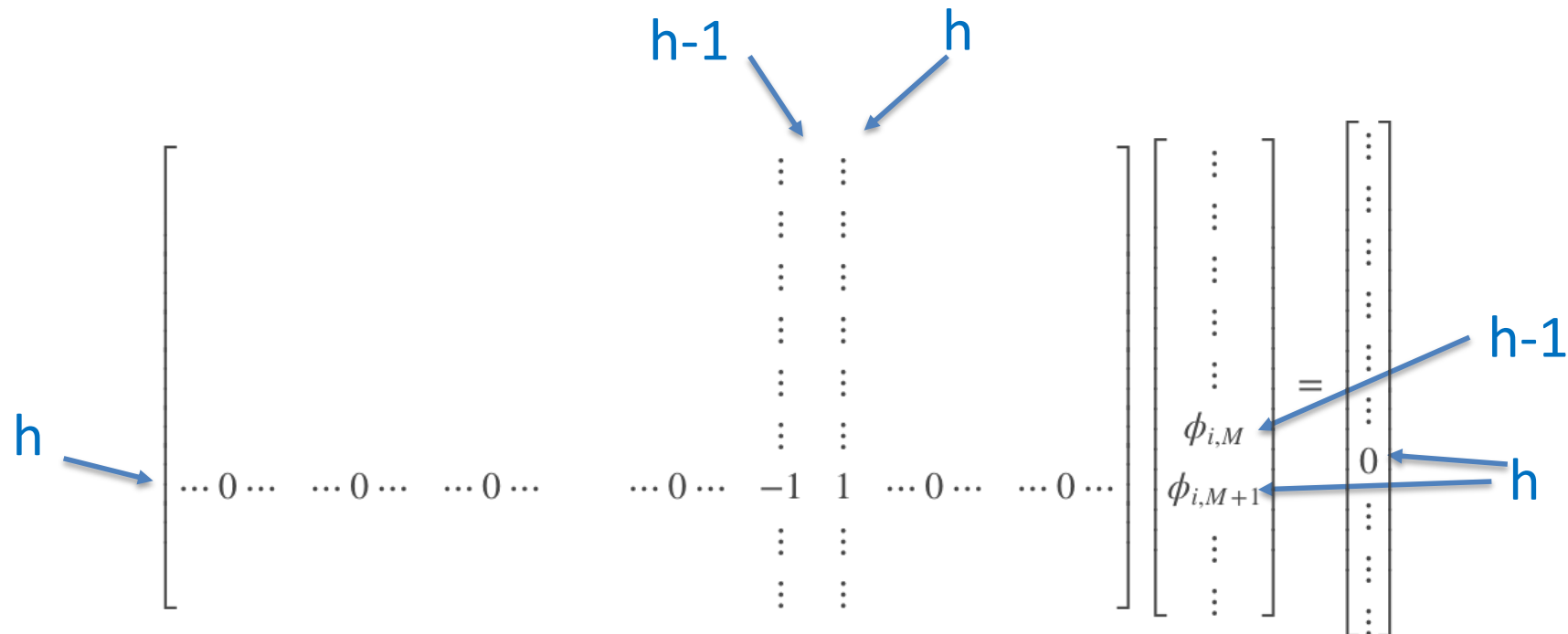


Example#2: finite difference implementation

- Zero flux on the x-axis

$$\begin{cases} \phi_{i,M+1} - \phi_{i,M} = 0 \\ j = M + 1 \end{cases}$$

```
elseif (j==1) %x-axis (thetaj=0)
    A(h,h)=1;
    A(h,h+1)=-1;|
elseif (j==M+1) %x-axis (thetaj=pi)
    A(h,h)=1;
    A(h,h-1)=-1;
```



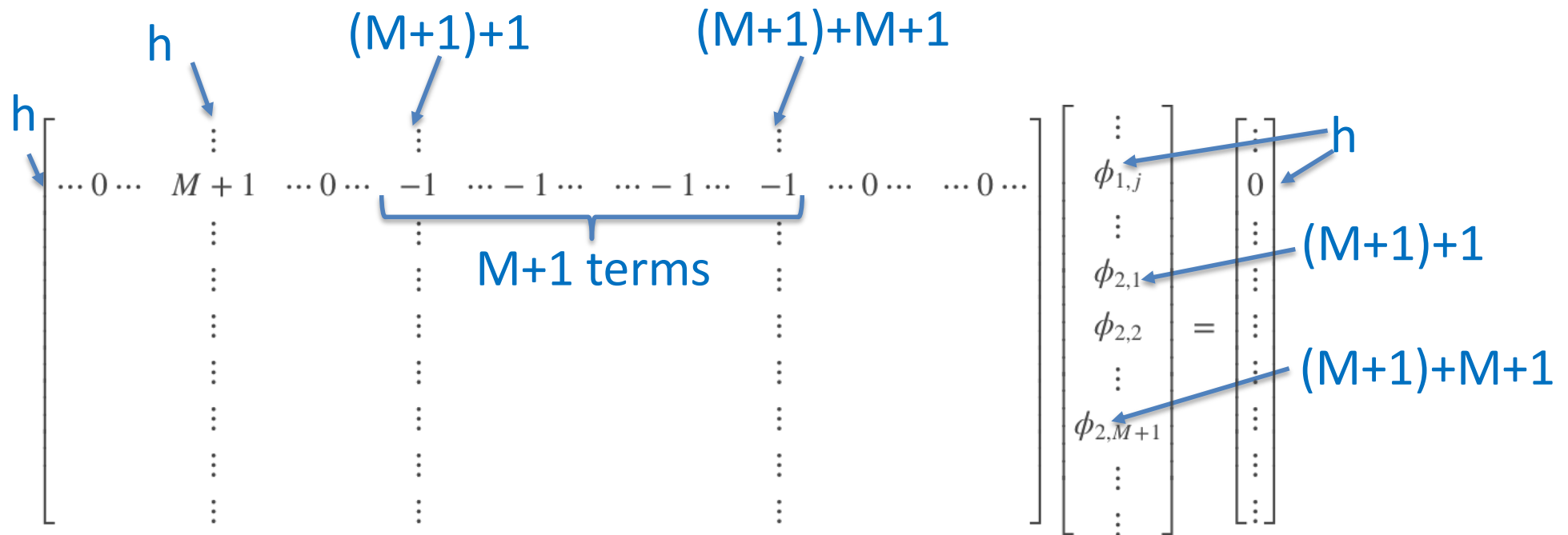
Example#2: finite difference implementation

- Center of the dish

$$\begin{cases} (M+1)\phi_{1,j} - \sum_{j=1}^{M+1} \phi_{2,j} = 0 \\ i = 1 \text{ Note: } h=j \end{cases}$$

$\forall j$

```
if (i==1) %center of the dish
    A(h,h)=M+1;
    A(h,1+M+1:1+M+1+M)=-1;
```



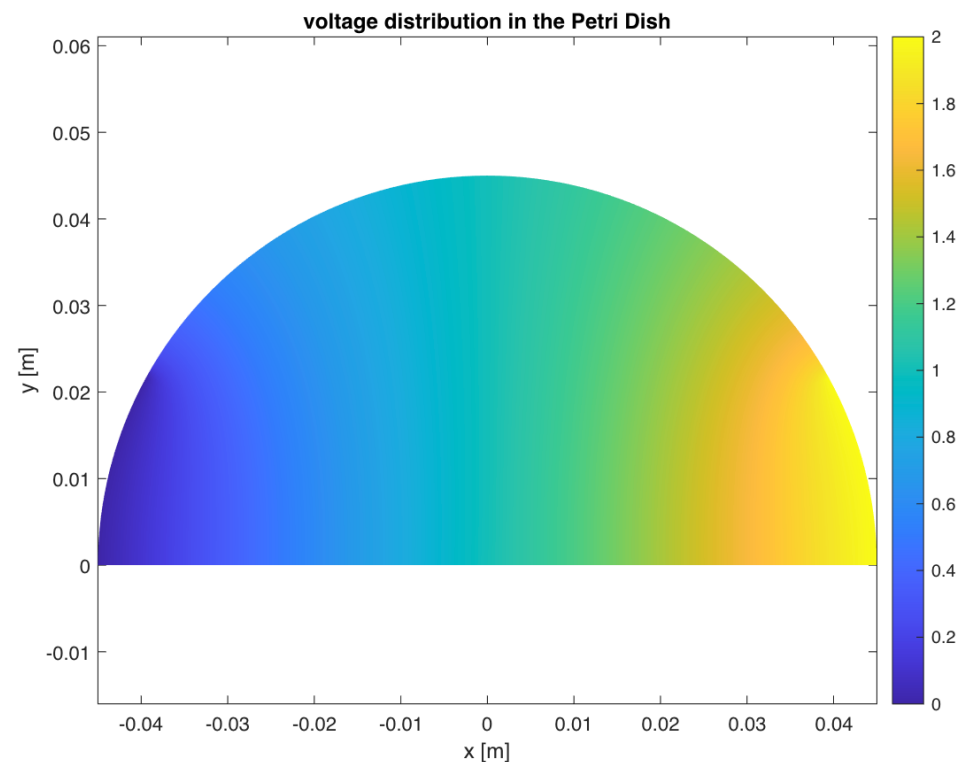
Example#2: solve and results

- Solve the system and format the data

```
%solve the system  
Y=A\B;  
V=reshape(Y,M+1,N+1);
```

- Plot results

```
[RR, th]=meshgrid(0:d_r:R,0:d_theta:pi);  
XX=RR.*cos(th);  
YY=RR.*sin(th);  
pcolor(XX,YY,V); colorbar; axis('equal'); shading('interp');  
xlabel('x [m]')  
ylabel('y [m]')  
title('voltage distribution in the Petri Dish');
```



Example#2: electric field

- Find $|\mathbf{E}|/|\mathbf{E}_c|$ ($\mathbf{E}_c$ is the electric field in the center of the dish)

$$\mathbf{E} = -\nabla\phi = -\begin{pmatrix} \frac{\partial\phi}{\partial x} \\ \frac{\partial\phi}{\partial y} \end{pmatrix} \quad \begin{array}{l} \longrightarrow \frac{\partial\phi}{\partial x} = \frac{x}{r} \frac{\partial\phi}{\partial r} - \frac{y}{r^2} \frac{\partial\phi}{\partial\theta} \\ \longrightarrow \frac{\partial\phi}{\partial y} = \frac{y}{r} \frac{\partial\phi}{\partial r} + \frac{x}{r^2} \frac{\partial\phi}{\partial\theta} \end{array} \quad \begin{array}{l} x = r \cos(\theta) \\ y = r \sin(\theta) \end{array}$$

$$|\mathbf{E}| = \sqrt{\left(\frac{\partial\phi}{\partial x}\right)^2 + \left(\frac{\partial\phi}{\partial y}\right)^2}$$

$$\mathbf{E}_c = -\begin{pmatrix} \left.\frac{\partial\phi}{\partial x}\right|_{x=0,y=0} \\ 0 \end{pmatrix}$$

Exploiting the
symmetry
about the x-axis

$$|\mathbf{E}_c| = \left|\left.\frac{\partial\phi}{\partial x}\right|_{x=0,y=0}\right| = \left|\left.\frac{\partial\phi}{\partial r}\right|_{r=0,\theta=0}\right|$$

Since the x-axis
is aligned with
the $\theta=0$ axis at
 $r=0$.

Example#2: electric field

- Region of uniform field

- We define a scalar function which is 1 when the electric field is within 10% of variation

$$\Gamma_A = \begin{cases} 1 & 0.9 \leq \frac{E}{E_c} \leq 1.1 \\ 0 & \text{otherwise} \end{cases}$$

- The area for which the electric field magnitude is +/- 10% of the value in the center is:

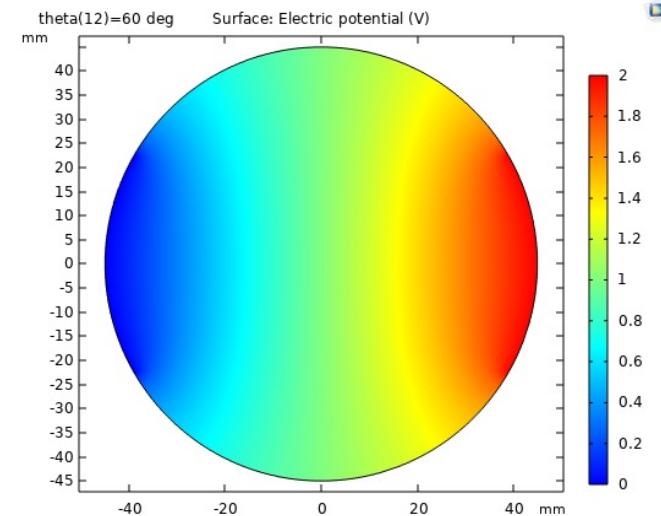
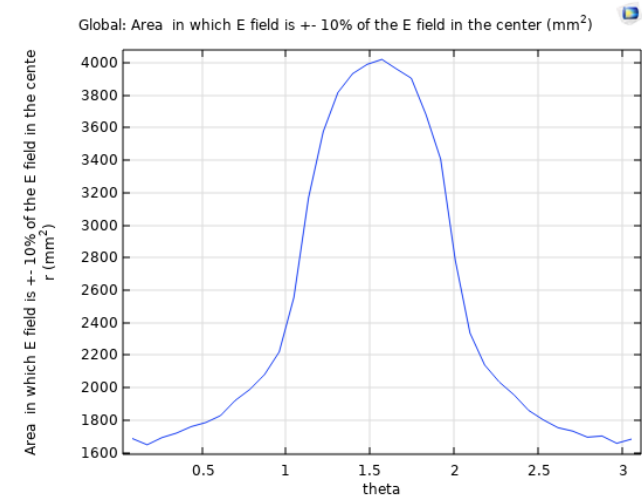
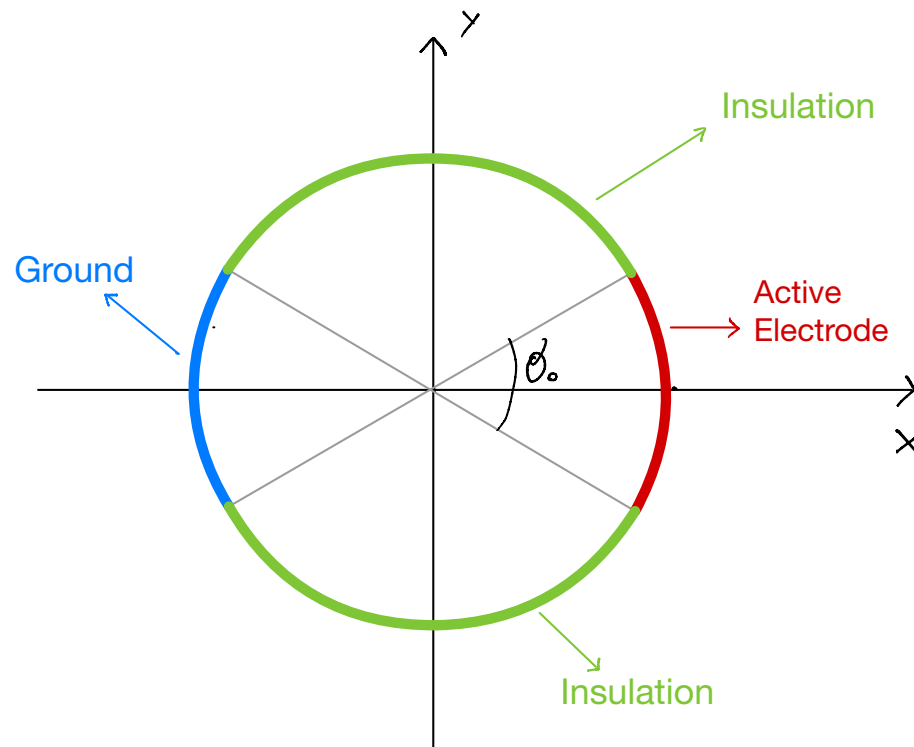
$$Area = \int_0^\pi \int_0^R \Gamma_A(r, \theta) r dr d\theta$$

- Evaluate the area against the electrode angle θ_0 (PetriDishUniformField.m)

How to solve the problem of the PetriDish
with in Comsol?

Example#2: Finite element solution in COMSOL

- Bidimensional problem: “PetriDishFE.mph”



Model of excitable tissues: neural, heart, muscular

- Excitability is a property of the cell membrane

Changes in space and time

intracellular potential

extracellular potential

Transmembrane potential $\rightarrow V_m = \phi_I - \phi_O$

Membrane current per unit of membrane area

Capacitive current per unit of membrane area

Membrane capacitance per unit of membrane area

$$I_m = I_C + I_{ion} = C_m \frac{\partial V_m}{\partial t} + I_{ion}$$

Different model depending on the specific tissue: neural (myelinated, unmyelinated), cardiac muscular

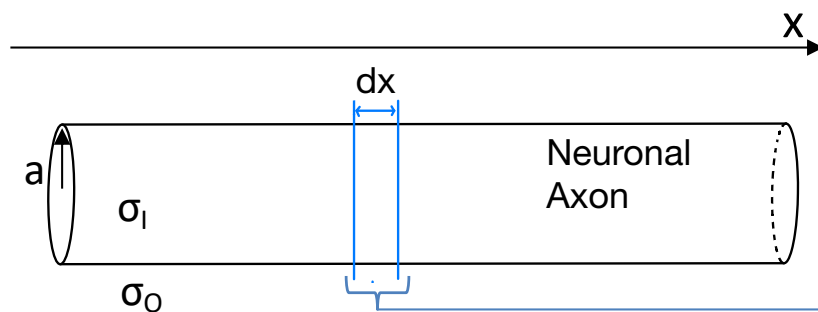
Ionic current per unit of membrane area

Hodgkin and Huxley model

- The H.H model describes the ionic mechanisms in unmyelinated neural tissue
- Based on electrophysiological experiments the H.H model describes the neural electrical impulse (i.e. the action potential)
- Most computational models of the electrical activity of nerve, muscle and heart are based on the formalism of Hodgkin and Huxley (bio-physical models)

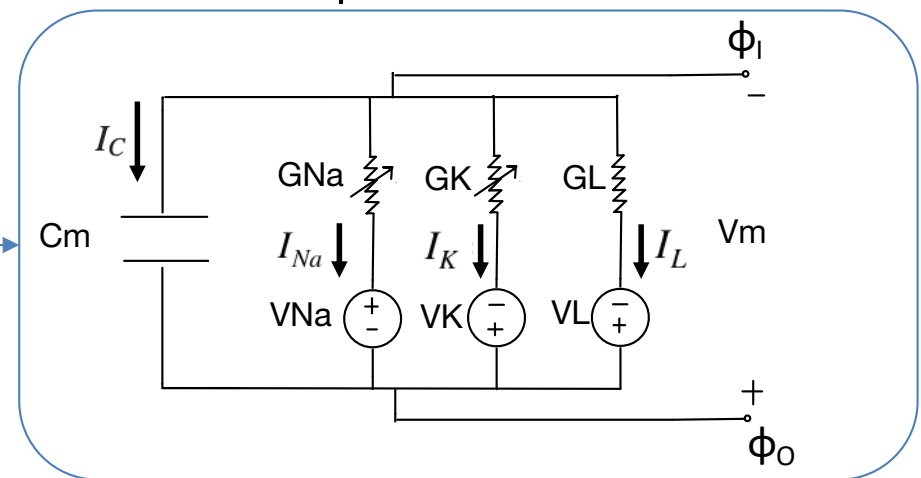
Hodgkin and Huxley model

- A small patch of the neuron membrane is described by a capacitance C_m in parallel with three conductance branches representing channels associated to transmembrane ionic currents.
- The patch of the neural cell membrane is represented as a capacitance C_m in parallel with conductance paths representing transmembrane channels for sodium ions (G_{Na}), potassium ions (G_K) and a non-specific leakage (G_L). Each conductance is in series with a voltage source, representing the equilibrium potential (or reversal potential) for that channel (according to Nernst law the equilibrium potentials of Sodium/Potassium branches depend on intra and extra cellular ionic concentration)
- G_{Na} and G_K are variable and follow a voltage-dependent kinetics .
- ϕ_i and ϕ_o denote the voltages inside (intracellular potential) and outside (extracellular potential) the neuron, with their difference equal to the transmembrane potential V_m



σ_i intracellular conductivity (conductivity of the axoplasm)

σ_o extracellular conductivity



Hodgkin and Huxley model

Ionic current per unit of
membrane area

$$I_{ion} = I_{Na} + I_K + I_L$$

$V_r \rightarrow$ membrane
rest potential

Sodium

$$I_{Na} = G_{Na}(V_m - V_{Na})$$

$$G_{Na} = \bar{G}_{Na} m^3 h$$

$$\frac{dm}{dt} = \alpha_m (1 - m) - \beta_m m$$

$$\frac{dh}{dt} = \alpha_h (1 - h) - \beta_h h$$

$$\alpha_m = \frac{0.1(25 - v_m)}{\exp[0.1(25 - v_m)] - 1}$$

$$\beta_m = 4 \exp\left(-\frac{v_m}{18}\right)$$

$$\alpha_h = 0.07 \exp\left(-\frac{v_m}{20}\right)$$

$$\beta_h = \left\{ \exp\left[\frac{(30 - v_m)}{10}\right] + 1 \right\}^{-1}$$

[msec⁻¹]

$$v_m = V_m - V_r$$

Potassium

$$I_K = G_K(V_m + V_K)$$

$$G_K = \bar{G}_K n^4$$

$$\frac{dn}{dt} = \alpha_n (1 - n) - \beta_n n$$

$$\alpha_n = \frac{0.01(10 - v_m)}{\left[\exp\left(\frac{10 - v_m}{10}\right) - 1\right]}$$

$$\beta_n = 0.125 \exp\left(\frac{-v_m}{80}\right)$$

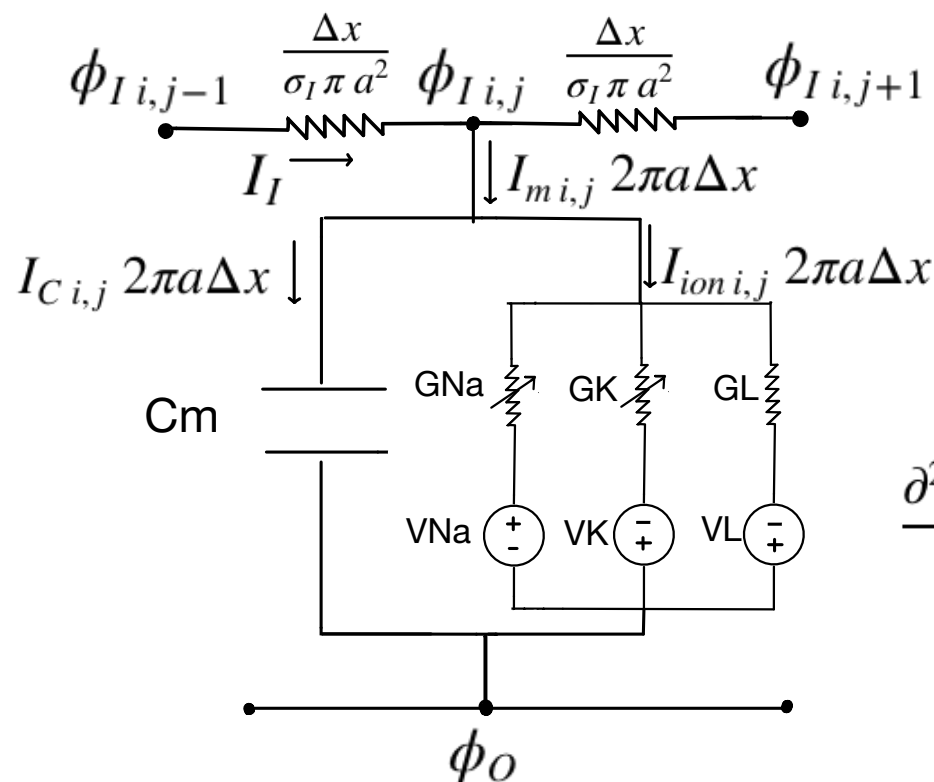
[msec⁻¹]

Leakage

$$I_L = G_L(V_m + V_L)$$

PDE describing the action potential propagation axon

- We discretize the x domain into nodes of distance Δx apart and we derive the PDE governing V_m in the limit as $\Delta x \rightarrow 0$
- We represent the electrical equivalent circuit of three adjacent nodes to derive the PDE



ϕ_O can be considered constant

$i \rightarrow$ temporal index $j \rightarrow$ spatial index

$$\left(\frac{2}{a \sigma_I}\right) I_{m i, j} = \frac{\phi_{I i, j+1} - 2 \phi_{I i, j} + \phi_{I i, j-1}}{\Delta x^2}$$

$$= \frac{V_{m i, j+1} - 2 V_{m i, j} + V_{m i, j-1}}{\Delta x^2}$$

$$\frac{\partial^2 V_m(t, x)}{\partial x^2} = \left(\frac{2}{a \sigma_I}\right) I_m = \left(\frac{2}{a \sigma_I}\right) \left[C_m \frac{\partial V_m}{\partial t} + I_{ion} \right]$$

$$\frac{\partial V_m}{\partial t} = \frac{1}{C_m} \left[\frac{a \sigma_I}{2} \frac{\partial^2 V_m(t, x)}{\partial x^2} - I_{ion} \right]$$

Example 3: Propagation of the action potential (AP)

- Consider a cylindrical axonal nerve membrane of radius a . Assume the extracellular potential to be constant. The membrane ionic current is governed by the Hodgkin-Huxley formulations. Derive:
 - The discretized PDE using FD approximation of derivatives
 - Solve the PDE for a cable length of 5 cm when a current pulse (ie) is delivered in $x = 1$ cm (characteristics of the pulse are reported in the picture below).
 - Simulation time: 150 msec
 - Plot the membrane potential on the axon for $t=30$ msec
 - Plot membrane potential at : $x=L/4$, $x=L/2$ and $x=3/4L$
 - Place an electrode ($x=3.5$ cm, $d=0.2$ cm) external to the axon and simulate the measured potential ($\sigma_o = 1 \text{ mS cm}^{-1}$)

$$C_m = 1 \mu\text{F cm}^{-2}$$

$$G_{Na} = 120 \text{ mS cm}^{-2}$$

$$G_K = 36 \text{ mS cm}^{-2}$$

$$G_L = 0.3 \text{ mS cm}^{-2}$$

$$a = 0.0025 \text{ cm}$$

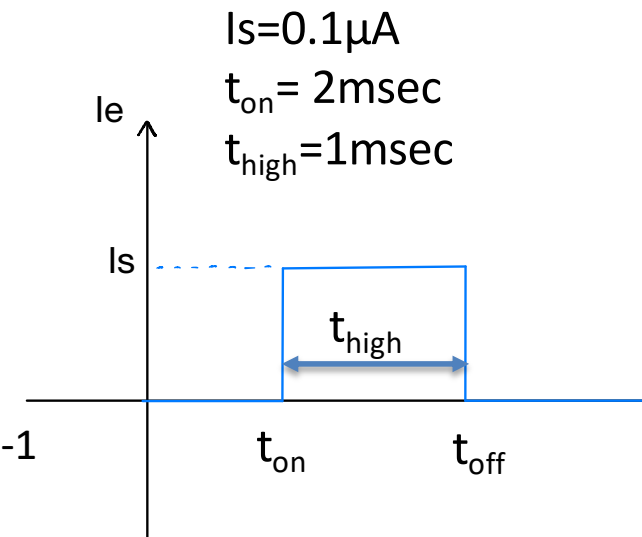
$$V_r = -60 \text{ mV}$$

$$V_{Na} = 55 \text{ mV}$$

$$V_K = 72 \text{ mV}$$

$$V_L = 49.387 \text{ mV}$$

$$\sigma_i = 1.6667 \text{ mS cm}^{-1}$$



Example 3: discretized PDE

$$\frac{\partial V_m}{\partial t} = \frac{1}{C_m} \left[\frac{a \sigma_I}{2} \frac{\partial^2 V_m(t, x)}{\partial x^2} - I_{ion} \right]$$

$$\Delta t = \frac{t_{sim}}{N} \quad \Delta x = \frac{L}{H}$$

$$t_i = (i - 1) \Delta t \quad i = 1 \dots N + 1$$

$$x_j = (j - 1) \Delta x \quad j = 1 \dots H + 1$$

The problem is time dependant

We define a grid over t and x

t_{sim} -> simulation time

L -> length of the axon

$$\frac{V_{mi+1,j} - V_{mi,j}}{\Delta t} = \frac{1}{C_m} \left[\frac{a \sigma_I}{2} \frac{V_{mi,j+1} - 2 V_{mi,j} + V_{mi,j-1}}{\Delta x^2} - I_{ioni,j} \right]$$

$$I_{ioni,j} = \overline{G}_{Na} m_{i,j}^3 h_{i,j} (V_{mi,j} - V_{Na}) + \overline{G}_K n_{i,j}^4 (V_{mi,j} + V_k) + G_L (V_{mi,j} + V_L)$$

$$\frac{m_{i+1,j} - m_{i,j}}{\Delta t} = \alpha_m(V_{mi,j}) (1 - m_{i,j}) - \beta_m(V_{mi,j}) m_{i,j}$$

$$\frac{n_{i+1,j} - n_{i,j}}{\Delta t} = \alpha_n(V_{mi,j}) (1 - n_{i,j}) - \beta_n(V_{mi,j}) n_{i,j}$$

$$\frac{h_{i+1,j} - h_{i,j}}{\Delta t} = \alpha_h(V_{mi,j}) (1 - h_{i,j}) - \beta_h(V_{mi,j}) h_{i,j}$$

Example 3: Initial conditions and boundary conditions

- Initial conditions
 - The axonal membrane is at rest for $t=0$ ($V_m=V_r$ in all the length of the axon)
- Zero flux boundary conditions
 - No currents flowing at each ends of the cable that represents the axon -> thus the current flowing in the intra cellular space (I_i) is zero at each cable ends

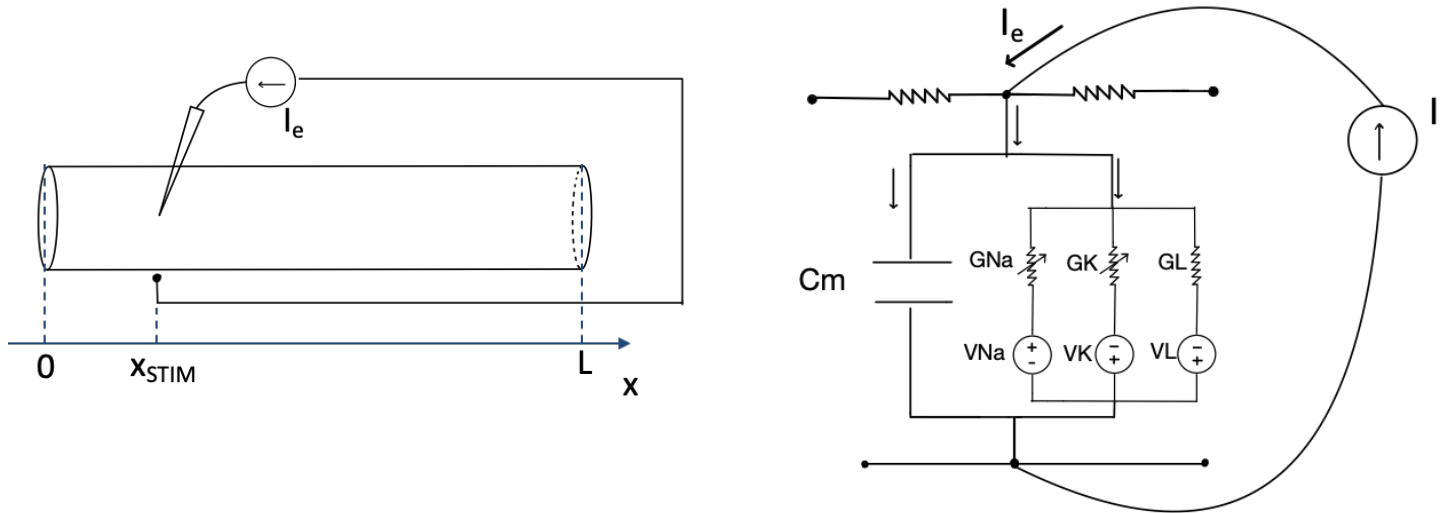
$$I_i|_{x=0} = 0 = \sigma_i \pi a^2 \frac{\phi_{i,1} - \phi_{i,2}}{\Delta x} = \sigma_i \pi a^2 \frac{V_{m,1} - V_{m,2}}{\Delta x} \Rightarrow V_{m,1} = V_{m,2}$$

$$I_i|_{x=L} = 0 = \sigma_i \pi a^2 \frac{\phi_{i,H} - \phi_{i,H+1}}{\Delta x} = \sigma_i \pi a^2 \frac{V_{m,H} - V_{m,H+1}}{\Delta x} \Rightarrow V_{m,H+1} = V_{m,H}$$

Example 3: current stimulation

Note: it can be considered
a boundary condition

- The discretized PDE changes in the spatial node in which the current is applied



From x_{STIM} we derive the index $j_S = \text{round}(x_{STIM}/\Delta x)$

$$\frac{V_{mi+1,j_S} - V_{mi,j_S}}{\Delta t} = \frac{1}{C_m} \left[\frac{a \sigma_I}{2} \frac{V_{mi,j_S+1} - 2V_{mi,j_S} + V_{mi,j_S-1}}{\Delta x^2} - I_{ion i,j_S} + \frac{I_{e i}}{2\pi a \Delta x} \right]$$

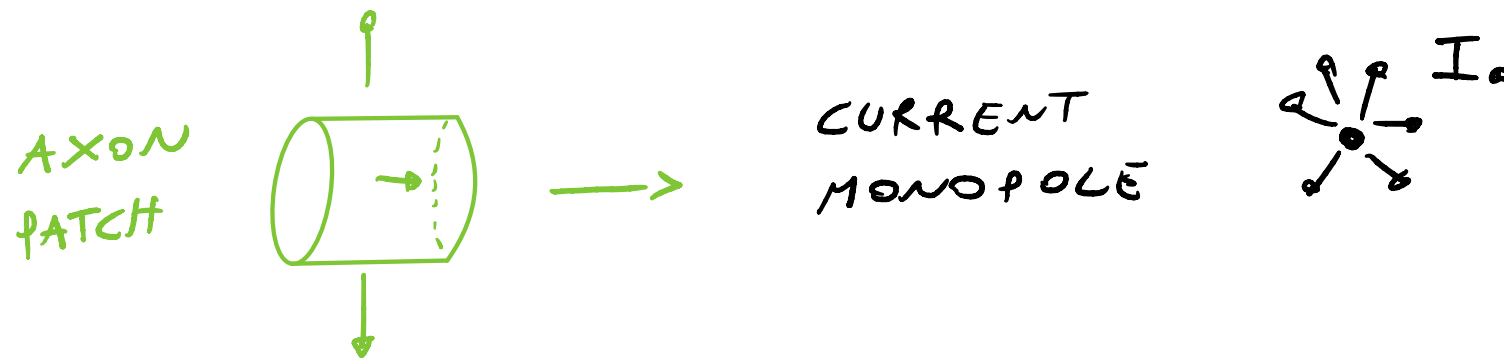
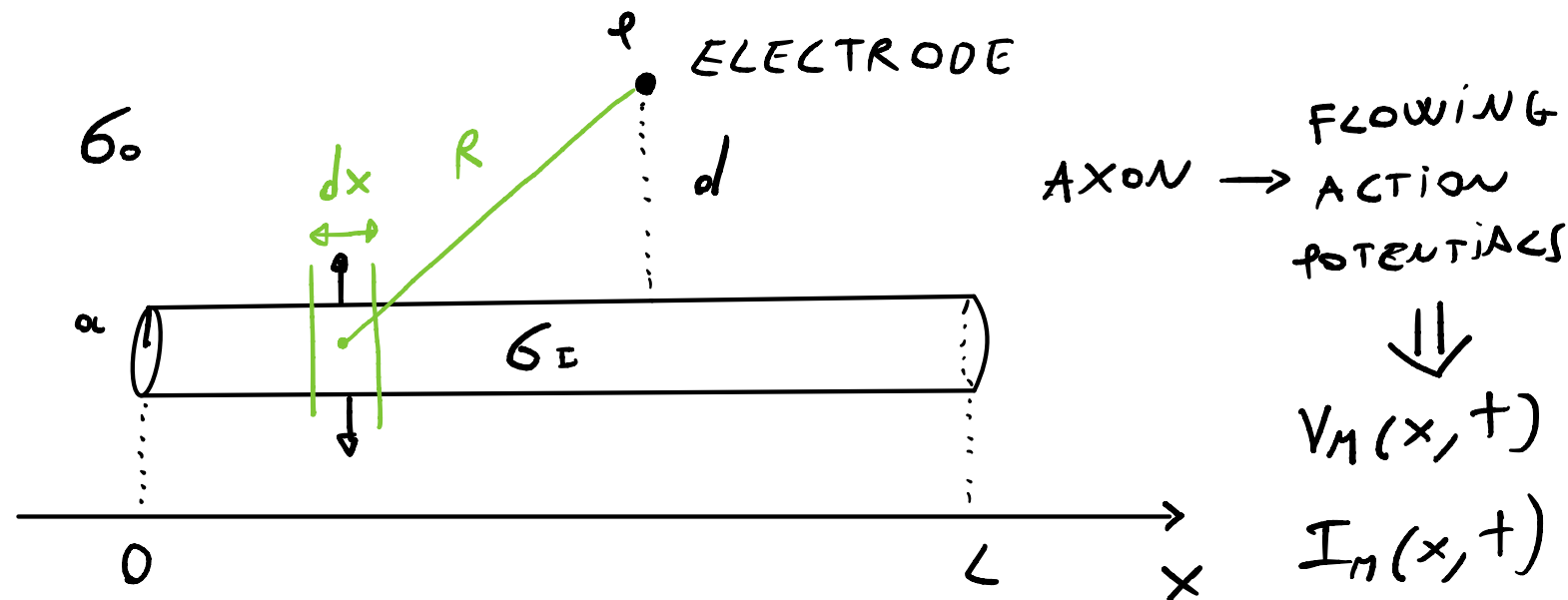
Example 3: solution with finite FD explicit method

Note: at each time step the only unknowns are the values of the state variables (V_m , m , h , n) at the next time step $\rightarrow$ if we know the initial conditions, the problem can be solved with an iterative method (FD equations that express one unknown in terms of known values)

$$\left\{ \begin{array}{l} V_{mi+1,j} = V_{mi,j} + \frac{\Delta t}{C_m} \left[\frac{a \sigma_I}{2} \frac{V_{mi,j+1} - 2V_{mi,j} + V_{mi,j-1}}{\Delta x^2} - I_{ioni,j} \right] \\ n_{i+1,j} = n_{i,j} + \Delta t [\alpha_n(V_{mi,j}) (1 - n_{i,j}) - \beta_n(V_{mi,j}) n_{i,j}] \\ m_{i+1,j} = m_{i,j} + \Delta t [\alpha_m(V_{mi,j}) (1 - m_{i,j}) - \beta_m(V_{mi,j}) m_{i,j}] \\ h_{i+1,j} = h_{i,j} + \Delta t [\alpha_h(V_{mi,j}) (1 - h_{i,j}) - \beta_h(V_{mi,j}) h_{i,j}] \end{array} \right.$$

Derive the explicit (iterative) method in Matlab! AxonPDE_explicit.m (example 3)

Extracellular potential induced by the action potential (1)



Extracellular potential induced by the action potential (2)

$$\phi_{\text{MONOPOL}} = \frac{I_0}{4\pi\sigma R} \quad \text{Diagram: A circle with a center point and a radius vector labeled } r. \quad \bar{J} = \frac{I_0}{4\pi R^2} \hat{r}$$

$\Rightarrow$ AXON PATCH $\longrightarrow$ CURRENT MONOPOL $\left(\frac{I_m 2\pi a dx}{[A]} \right)$

$$\Rightarrow \phi_p = \frac{1}{4\pi\sigma} \int_0^L \frac{I_m 2\pi a}{R} dx$$

$$R = \sqrt{d^2 + (x - x_p)^2}$$

Example 3: solution with finite FD explicit method

Check consistency of the units:

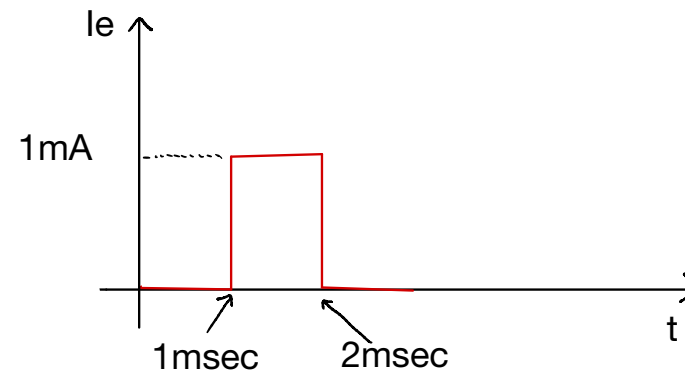
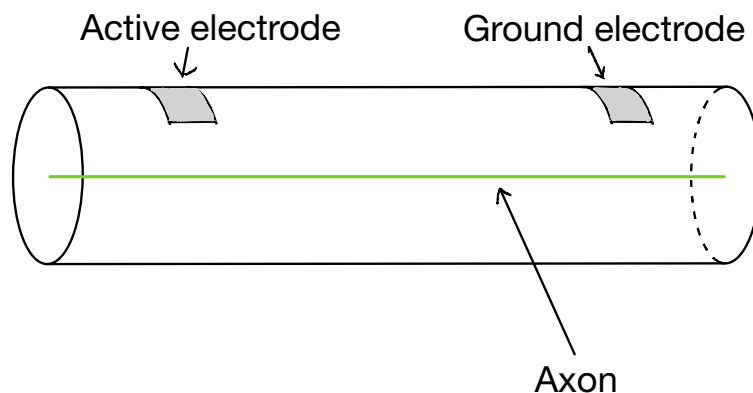
- Potentials -> mV
- Ionic/capacitive currents -> $\mu\text{A cm}^{-2}$
- Current stimulus -> μA
- C_m -> $\mu\text{F cm}^{-2}$
- Conductance -> mS cm^{-1}
- Time -> msec
- Space -> cm
- $\frac{\Delta t a \sigma_I}{C_m}$ -> cm^2 ok!
- $\frac{\Delta t}{C_m} I_{ion}$ -> mV ok!

It can be demonstrated that the necessary condition for stability of this explicit method method is:

$$D_i \frac{\Delta t}{\Delta x^2} = \left(\frac{a \sigma_I}{2 C_m} \right) \frac{\Delta t}{\Delta x^2} \leq \frac{1}{2}$$

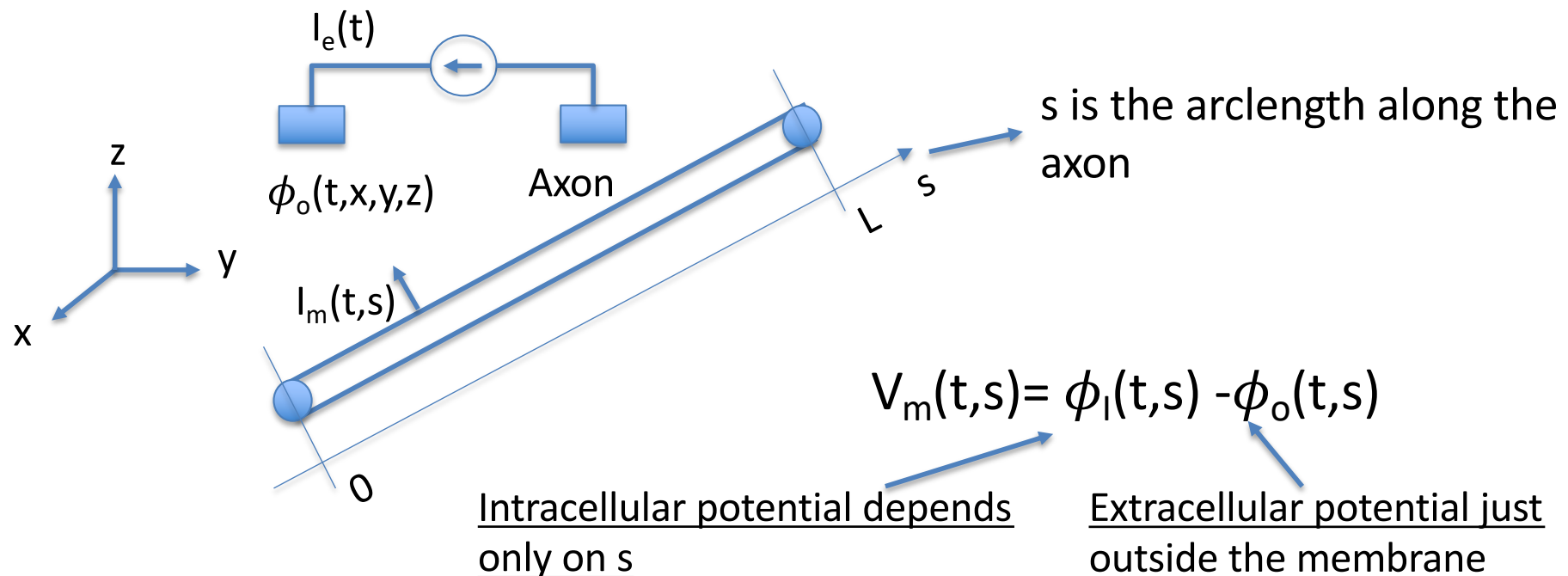
Example 4: Axonal stimulation using nerve cuff electrodes

- Consider a $L=10$ mm length nerve bundle of radius $R=1$ mm.
- The bundle is stimulated by two cuff electrodes represented as squares of side length 1 mm wrapped around the bundle and having a center spacing of 7 mm along the main axis of the bundle
- The active left cuff electrode delivers an inward pulse current of 0.5mA over 1msec, the right electrode is held at ground. Considering an axon in the center of the bundle (aligned with the x-axis), derive:
 1. The PDE governing intra- and extracellular potentials on the axon
 2. Solve in FE for V_m
 3. Plot the membrane potential over axon length at different times (1msec, 3msec, 5msec, 7msec, 9msec)
 4. Plot the external potential in the point $(0.75*L, -0.55*R, 0)$, animate results

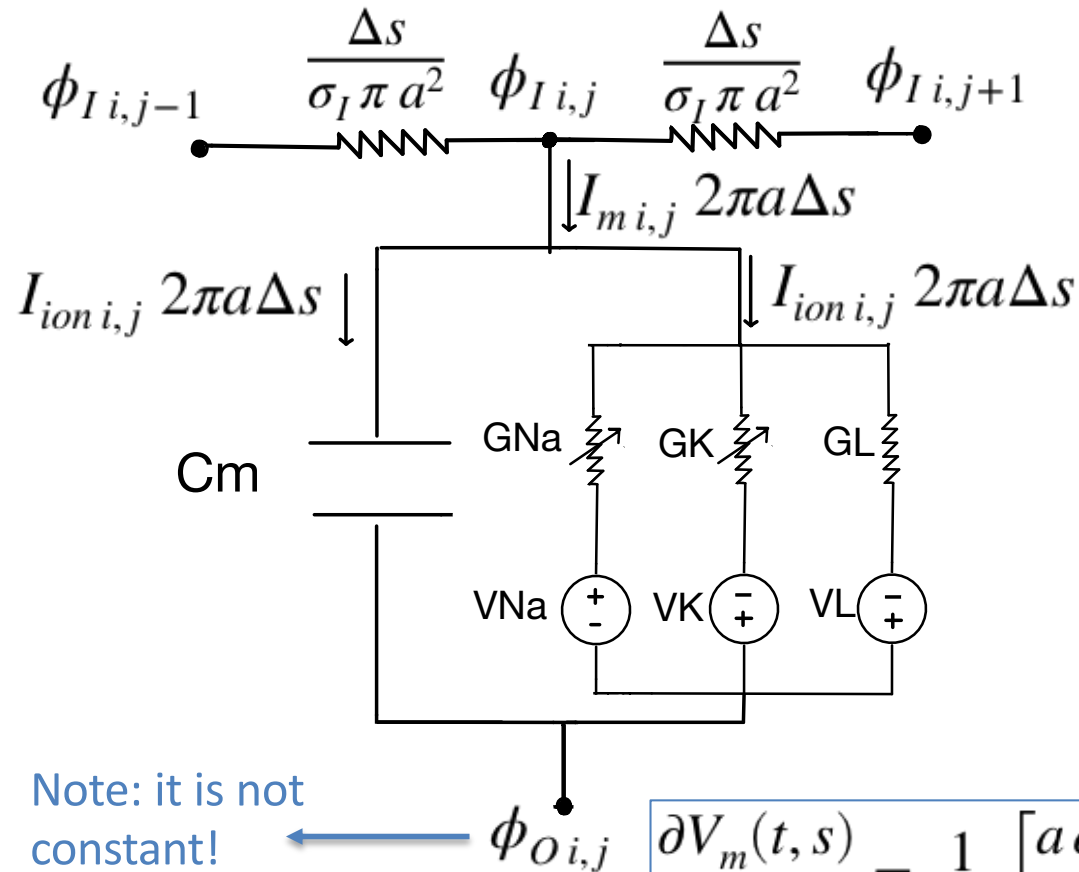


PDE governing intra- and extracellular potentials on the axon

- Same as the AP propagation, but we can not consider a constant extracellular potential
- The extracellular potential is modified according to the current delivered by the two electrodes



The PDE governing intra- and extracellular potentials on the axon

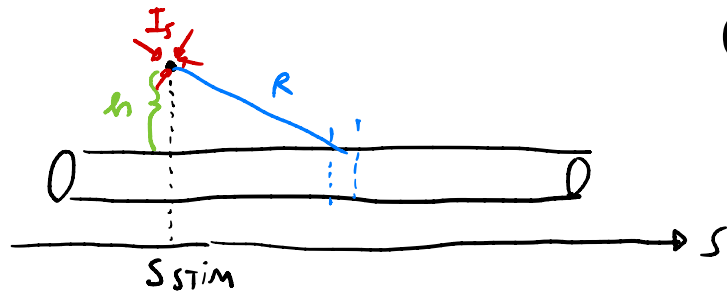


This term describes the effect of the stimulus (i.e., current delivered in the extracellular space)

$$\frac{\partial V_m(t, s)}{\partial t} = \frac{1}{C_m} \left[\frac{a \sigma_I}{2} \frac{\partial^2 V_m(t, s)}{\partial s^2} + \frac{a \sigma_I}{2} \frac{\partial^2 \phi_O(t, s)}{\partial s^2} - I_{ion} \right]$$

Hodgkin and Huxley model

Example 5: Axon stimulation with monopolar electrode



$$I_s = 8e3 \text{ uA}$$

$$\Delta t_s = 1 \text{ msec}$$

$$d = 0.6 \text{ cm}$$

$$a = 0.005 \text{ cm}$$

All other Axon parameters from example 4

$$s_{\text{stim}} = 1 \text{ cm}$$

$$F = \frac{\int \phi^2 ds}{\int ds^2} \rightarrow \text{ACTIVATION FUNCTION}$$

① WHY?

$$F > 0$$

NECESSARY CONDITION FOR STIMULATION

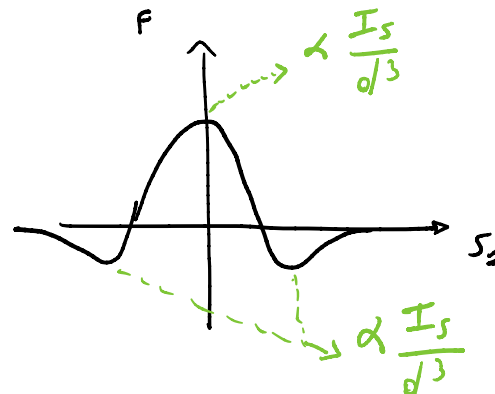
② WHY NOT SUFFICIENT?

MONOPOLAR CURRENTS $\phi_0 = \frac{1}{4\pi\epsilon_0} \frac{-I_s}{R}$

$$R = \sqrt{h^2 + (s - s_{\text{stim}})^2}$$

$$F = \frac{I_s}{4\pi\epsilon_0} \frac{(d^2 - 2s_1^2)}{(d^2 + s_1^2)^{5/2}}$$

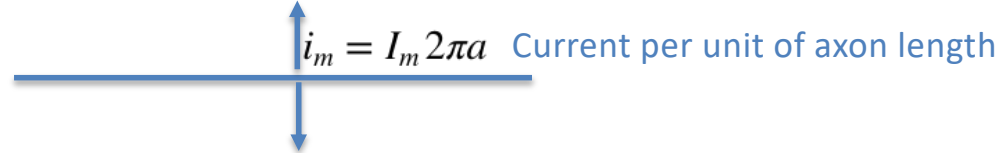
$$(s - s_{\text{stim}}) = s_1$$



Example 5#:
Axon_PDEFesstim.m

Nerve cuff stimulation: FE solution in Comsol

- We model the axon as a lower dimensional edge (i.e., a line) in a 3D domain
- The 3D domain is the nerve bundle and is modelled with the standard “electric currents” physics in which the dependent variable is the extracellular potential $\phi_o(t,x,y,z)$
 - The 3D domain describes the modification of the extracellular potential due to the current delivered by the cuff electrodes
 - In turn, the 3D domain is coupled with the axon at the axon boundary. Indeed, the axon deliver a current $I_m(t,s)$ in the extracellular space (line current source in COMSOL)



- The active electrode will be modelled as a floating potential (current $-I_e$, so entering in the electrode); the other electrode is held at ground
- The axon is modelled as a line in which $V_m(t,s)$ is governed by the PDE described in the previous slide
 - Note: the coupling between the axon and the 3D extracellular domain is represented by the the second spatial derivative of the extracellular potential

Nerve cuff stimulation: FE solution in Comsol – modelling the axon as an edge

- We arrange the axon PDE according to the Comsol PDE general form $e_a \frac{\partial^2 u}{\partial t^2} + d_a \frac{\partial u}{\partial t} + \nabla \cdot \Gamma = f$

$$e_a=0 \quad \underbrace{\left(\frac{2C_m}{a}\right)}_{d_a} \frac{\partial V_m(t,s)}{\partial t} + \frac{\partial}{\partial s} \left[\underbrace{-\sigma_I \frac{\partial V_m(t,s)}{\partial s} - \sigma_I \frac{\partial \phi_o(t,s)}{\partial s}}_{\Gamma} \right] = \underbrace{\left(-\frac{2}{a} I_{ion}\right)}_f$$

General Form
Edge PDE in
Comsol

$$\Gamma = -\sigma_I \left(\frac{\partial V_m}{\partial s} + \frac{\partial \phi_o}{\partial s} \right) \hat{s}$$

$$\frac{\partial U}{\partial s} = \nabla U \cdot \hat{s} = \underbrace{\frac{\partial U}{\partial x} \hat{x} \cdot \hat{s}}_{uT_x} + \underbrace{\frac{\partial U}{\partial y} \hat{y} \cdot \hat{s}}_{uT_y} + \underbrace{\frac{\partial U}{\partial z} \hat{z} \cdot \hat{s}}_{uT_z}$$

Note: on a lower dimensional boundary, such as the edge representing the axon, standard Comsol spatial derivatives are not defined, instead Comsol defines tangential derivatives (uT_x , uT_y , uT_z)

Nerve cuff stimulation: Built-in Comsol tangential derivatives

$$\left(\frac{2C_m}{a}\right) \frac{\partial V_M}{\partial t} + \left[-\sigma_T \frac{\partial^2 V_M}{\partial s^2} - \sigma_T \frac{\partial^2 \phi_0}{\partial s^2} \right] = \left(-\frac{2}{a} I_{ion} \right)$$

$$\left(\frac{2C_m}{a}\right) \frac{\partial V_M}{\partial t} + \frac{\partial}{\partial s} \left[-\sigma_T \frac{\partial V_M}{\partial s} - \sigma_T \frac{\partial \phi_0}{\partial s} \right] = \left(-\frac{2}{a} I_{ion} \right)$$

$$\vec{F} = -\sigma_T \left[\frac{\partial V_M}{\partial s} + \frac{\partial \phi_0}{\partial s} \right] \hat{s}$$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial s} F_s =$$

Nerve cuff stimulation: FE solution in Comsol – modelling the axon as an edge

- We arrange the gating variables ODE according to the comsol PDE general form

$$e_a \frac{\partial^2 u}{\partial u^2} + d_a \frac{\partial u}{\partial u} + \nabla \cdot \Gamma = f$$

General Form
Edge PDE in
Comsol

$$\mathbf{e}_a \frac{\partial^2 \mathbf{u}}{\partial t^2} + \mathbf{d}_a \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \Gamma = \mathbf{f} \quad \mathbf{u} = [n \ m \ h]^T$$

$$\mathbf{e}_a = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \mathbf{d}_a = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{f} = \begin{bmatrix} \alpha_n(1-n) - \beta_n n \\ \alpha_m(1-m) - \beta_m m \\ \alpha_h(1-h) - \beta_h h \end{bmatrix}$$

NerveCuffStimulation.mph (example 4)